

GET 321

Engineering Mathematics IV

Marking Scheme for the Theory Questions

Questions 2 to 7 · Types 1 to 5

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Contents

Type 1.....	3
Question 2: Differential Equations and Free Vibration.....	3
Question 3: RLC Circuit.....	4
Question 4: Series Solutions and Bessel's Equation.....	6
Question 5: Eigenvalues and Eigenvectors.....	7
Question 6: Legendre's Equation and the Wave Equation.....	9
Question 7: Numerical Differentiation.....	11
Type 2.....	14
Question 2: Differential Equations and Damped Vibration.....	14
Question 3: RLC Circuit.....	15
Question 4: Series Solutions and Bessel's Equation.....	16
Question 5: Eigenvalues and Eigenvectors.....	18
Question 6: Numerical Differentiation.....	19
Question 7: Legendre's Equation and the Wave Equation.....	21
Type 3.....	24
Question 2: Numerical Differentiation.....	24
Question 3: RLC Circuit.....	26
Question 4: Series Solutions and Bessel's Equation.....	27
Question 5: Legendre's Equation and the Wave Equation.....	29
Question 6: Eigenvalues and Eigenvectors.....	31
Question 7: Differential Equations and Free Vibration.....	32
Type 4.....	35
Question 2: Differential Equations and Free Vibration.....	35
Question 3: Numerical Differentiation.....	36
Question 4: Series Solutions and Bessel's Equation.....	38
Question 5: Eigenvalues and Eigenvectors.....	40
Question 6: Legendre's Equation and the Wave Equation.....	42
Question 7: RLC Circuit.....	44
Type 5.....	46
Question 2: Differential Equations and Free Vibration.....	46
Question 3: RLC Circuit.....	47
Question 4: Numerical Differentiation.....	49
Question 5: Eigenvalues and Eigenvectors.....	51
Question 6: Series Solutions and Bessel's Equation.....	52
Question 7: Legendre's Equation and the Wave Equation.....	54

Type 1

Question 2: Differential Equations and Free Vibration

(a) Non-homogeneous second-order ODE

Question 2(a). Find the solution to the engineering problem represented by the differential equation below. (6 marks)

$$\frac{d^2 y}{dx^2} - 4 \frac{dy}{dx} + 4y = x^2 + x^3 e^{2x} + \sin 2x$$

Complementary function. The auxiliary equation is

$$m^2 - 4m + 4 = (m - 2)^2 = 0 \Rightarrow m = 2 \text{ (repeated)}$$

$$y_c = (A + Bx)e^{2x}$$

Particular integral. Treat each term on the right separately: $y_p = y_1 + y_2 + y_3$.

For x^2 . Try $y_1 = ax^2 + bx + c$, so $y_1' = 2ax + b$ and $y_1'' = 2a$:

$$2a - 4(2ax + b) + 4(ax^2 + bx + c) = 4ax^2 + (4b - 8a)x + (2a - 4b + 4c) = x^2$$

$$4a = 1, 4b - 8a = 0, 2a - 4b + 4c = 0 \Rightarrow a = 1/4, b = 1/2, c = 3/8$$

For $x^3 e^{2x}$. Here e^{2x} is part of y_c (twice over), so put $y_2 = u(x)e^{2x}$. Then $y_2' = (u' + 2u)e^{2x}$, $y_2'' = (u'' + 4u' + 4u)e^{2x}$ and

$$y_2'' - 4y_2' + 4y_2 = [u'' + 4u' + 4u - 4u' - 8u + 4u]e^{2x} = u''e^{2x}$$

So $u'' = x^3$, giving $u = \frac{x^5}{20}$ and $y_2 = \frac{x^5}{20}e^{2x}$.

For $\sin 2x$. Try $y_3 = C \cos 2x + D \sin 2x$, so $y_3'' = -4y_3$:

$$-4y_3 - 4y_3' + 4y_3 = -4(-2C \sin 2x + 2D \cos 2x) = 8C \sin 2x - 8D \cos 2x = \sin 2x$$

so $C = 1/8$, $D = 0$ and $y_3 = 1/8 \cos 2x$.

$$\text{Answer. } y = (A + Bx)e^{2x} + \frac{x^5}{20}e^{2x} + \frac{x^2}{4} + \frac{x}{2} + \frac{3}{8} + \frac{1}{8} \cos 2x$$

(b) Undamped spring-mass system

Question 2(b). A spring with a mass of 2000 g has natural length of one-half a meter. A force of 25.6 N is required to maintain it stretched to a length of 700 mm. If the spring is stretched to a length of 700 mm and then released with initial velocity 0 m/s: (i) What will be the position of the mass at any time, t ? (ii) What length of travel (in m) would the mass achieve in $t = \pi/24$ seconds? (6 marks)

Spring constant. Hooke's law, $F=kx$, with the extension $x=0.7-0.5=0.2$ m:

$$k(0.2)=25.6 \Rightarrow k=128 \text{ N/m}$$

The mass is $m=2000 \text{ g}=2 \text{ kg}$. Let $x(t)$ be the displacement of the mass from equilibrium (the natural length, 0.5 m), positive in the direction of stretching.

Equation of motion. There is no damping, so

$$m \frac{d^2 x}{dt^2} + kx = 0 \Rightarrow 2 \frac{d^2 x}{dt^2} + 128x = 0 \Rightarrow \frac{d^2 x}{dt^2} + 64x = 0$$

The auxiliary equation $r^2+64=0$ gives $r=\pm 8i$, so

$$x(t)=c_1 \cos 8t + c_2 \sin 8t$$

Initial conditions. The spring starts stretched to 0.7 m, so $x(0)=0.2$, and it is released from rest, so $x'(0)=0$.

$$x(0)=c_1=0.2, x'(0)=8c_2=0 \Rightarrow c_2=0$$

(i) Position at any time.

$$x(t)=0.2 \cos 8t \text{ m}$$

The mass oscillates about the equilibrium position with amplitude 0.2 m, angular frequency $\omega=8 \text{ rad/s}$ and period $T=2\pi/8=\pi/4 \text{ s}$.

(ii) At $t=\pi/24 \text{ s}$.

$$x\left(\frac{\pi}{24}\right)=0.2 \cos\left(\frac{8\pi}{24}\right)=0.2 \cos \frac{\pi}{3}=0.2 \times 0.5=0.1 \text{ m}$$

The velocity $x'(t)=-1.6 \sin 8t$ stays negative for $0 < t < \pi/8$, and $\pi/24 < \pi/8$, so the mass moves steadily back towards equilibrium without turning. The distance travelled from the start is therefore

$$0.2 - 0.1 = 0.1 \text{ m}$$

and the spring's length at that moment is $0.5+0.1=0.6 \text{ m}$.

Answer. (i) $x(t)=0.2 \cos 8t \text{ m}$. (ii) At $t=\pi/24 \text{ s}$ the mass is 0.1 m from equilibrium, having travelled 0.1 m from its starting point (spring length 0.6 m).

Question 3: RLC Circuit

Question 3. A series circuit comprises a resistor of value 20Ω with a 1 H inductor and a capacitor whose capacitance is $2000 \mu\text{F}$. If the initial charge and current are both zero, find the charge at time t in these two cases: (i) the circuit is connected to a 12-volt battery; (ii) the circuit is connected to a generator producing a voltage of $E(t)=12 \sin 10t$. (6 marks) Find also the current in each case in (i) and (ii). (6 marks)

The circuit equation. By Kirchhoff's voltage law, with $I=dq/dt$,

$$L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{q}{C} = E(t)$$

Here $R=20\ \Omega$, $L=1\ \text{H}$ and $C=2000\ \mu\text{F}=2 \times 10^{-3}\ \text{F}$, so $1/C=500$:

$$\frac{d^2 q}{dt^2} + 20 \frac{dq}{dt} + 500 q = E(t), q(0)=0, I(0)=q'(0)=0$$

Complementary function. $r^2 + 20r + 500 = 0$ gives $r = -10 \pm \sqrt{100 - 500} = -10 \pm 20i$:

$$q_c = e^{-10t} (c_1 \cos 20t + c_2 \sin 20t)$$

Case (i): 12 V battery, $E=12$

Particular integral. Try a constant: $500 q_p = 12$, so $q_p = \frac{12}{500} = 0.024$.

$$q = e^{-10t} (c_1 \cos 20t + c_2 \sin 20t) + 0.024$$

$$q(0)=0: c_1 + 0.024 = 0 \Rightarrow c_1 = -0.024.$$

$$q'(0)=0: -10c_1 + 20c_2 = 0 \Rightarrow c_2 = 1/2 c_1 = -0.012.$$

$$q(t) = 0.024 - e^{-10t} (0.024 \cos 20t + 0.012 \sin 20t) = \frac{3}{250} [2 - e^{-10t} (2 \cos 20t + \sin 20t)]$$

Current. Differentiating,

$$I = \frac{dq}{dt} = 10 e^{-10t} (0.024 \cos 20t + 0.012 \sin 20t) - e^{-10t} (-0.48 \sin 20t + 0.24 \cos 20t) = 0.6 e^{-10t} \sin 20t$$

Case (ii): generator, $E=12 \sin 10t$

Particular integral. Try $q_p = A \cos 10t + B \sin 10t$, so $q_p'' = -100 q_p$:

$$(500 - 100)(A \cos 10t + B \sin 10t) + 20(-10A \sin 10t + 10B \cos 10t) = 12 \sin 10t$$

$$\cos 10t: 400A + 200B = 0, \sin 10t: 400B - 200A = 12$$

From the first, $B = -2A$; then $-800A - 200A = 12$, so $A = -0.012$ and $B = 0.024$:

$$q_p = -0.012 \cos 10t + 0.024 \sin 10t$$

Constants. $q(0) = c_1 - 0.012 = 0 \Rightarrow c_1 = 0.012$.

$$q'(0) = -10c_1 + 20c_2 + 10(0.024) = -0.12 + 20c_2 + 0.24 = 0 \Rightarrow c_2 = -0.006.$$

$$q(t) = e^{-10t} (0.012 \cos 20t - 0.006 \sin 20t) - 0.012 \cos 10t + 0.024 \sin 10t$$

Current.

$$I = \frac{dq}{dt} = -e^{-10t} (0.24 \cos 20t + 0.18 \sin 20t) + 0.24 \cos 10t + 0.12 \sin 10t$$

Check: $I(0) = -0.24 + 0.24 = 0$. ✓ The e^{-10t} terms are the transient, which dies away; the steady-state current is $0.24 \cos 10t + 0.12 \sin 10t$.

Answer. (i) $q = 0.024 - e^{-10t}(0.024 \cos 20t + 0.012 \sin 20t)$ C and $I = 0.6e^{-10t} \sin 20t$ A.
(ii) $q = e^{-10t}(0.012 \cos 20t - 0.006 \sin 20t) - 0.012 \cos 10t + 0.024 \sin 10t$ C and
 $I = -e^{-10t}(0.24 \cos 20t + 0.18 \sin 20t) + 0.24 \cos 10t + 0.12 \sin 10t$ A.

Question 4: Series Solutions and Bessel's Equation

(a) Power series solution

Question 4(a). Find the power series solution of $y'' + y = 0$. (6 marks)

Assume a power series solution about $x=0$:

$$y = \sum_{n=0}^{\infty} a_n x^n, y'' = \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} = \sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2} x^n$$

Substituting into $y'' + y = 0$:

$$\sum_{n=0}^{\infty} [(n+2)(n+1)a_{n+2} + a_n] x^n = 0$$

Every coefficient must vanish, which gives the **recurrence relation**

$$a_{n+2} = -\frac{a_n}{(n+2)(n+1)}, n=0, 1, 2, \dots$$

Even coefficients (in terms of a_0):

$$a_2 = -\frac{a_0}{2!}, a_4 = -\frac{a_2}{4 \cdot 3} = \frac{a_0}{4!}, a_6 = -\frac{a_4}{6 \cdot 5} = -\frac{a_0}{6!}, \dots$$

Odd coefficients (in terms of a_1):

$$a_3 = -\frac{a_1}{3!}, a_5 = -\frac{a_3}{5 \cdot 4} = \frac{a_1}{5!}, a_7 = -\frac{a_1}{7!}, \dots$$

Hence

$$y = a_0 \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots \right) + a_1 \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots \right)$$

The two series are the Maclaurin series of $\cos x$ and $\sin x$, and both converge for all x .

Answer. $y = a_0 \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots \right) + a_1 \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots \right) = a_0 \cos x + a_1 \sin x$, where a_0 and a_1 are arbitrary constants.

(b) Bessel's equation

Question 4(b). Find the general solution, in terms of Bessel functions, of the equation $xy'' + y' + \left(\frac{1}{x}\right)y = 0$. (6 marks)

The fraction in the equation reduces to $1/2$, so the equation is

$$x y'' + y' + 1/2 y = 0$$

Change of variable. Equations of the form $x y'' + y' + \lambda y = 0$ become Bessel's equation under $z = 2\sqrt{\lambda x}$. Here $\lambda = 1/2$, so put

$$z = 2\sqrt{x/2} = \sqrt{2x}, z^2 = 2x, \frac{dz}{dx} = \frac{1}{z}$$

Then

$$\frac{dy}{dx} = \frac{1}{z} \frac{dy}{dz}, \frac{d^2y}{dx^2} = \frac{1}{z} \frac{d}{dz} \left(\frac{1}{z} \frac{dy}{dz} \right) = \frac{1}{z^2} \frac{d^2y}{dz^2} - \frac{1}{z^3} \frac{dy}{dz}$$

With $x = z^2/2$:

$$x \frac{d^2y}{dx^2} + \frac{dy}{dx} + \frac{1}{2} y = \frac{1}{2} \frac{d^2y}{dz^2} - \frac{1}{2z} \frac{dy}{dz} + \frac{1}{z} \frac{dy}{dz} + \frac{1}{2} y = \frac{1}{2} \left(\frac{d^2y}{dz^2} + \frac{1}{z} \frac{dy}{dz} + y \right) = 0$$

Multiplying by $2z^2$:

$$z^2 \frac{d^2y}{dz^2} + z \frac{dy}{dz} + (z^2 - \nu^2) y = 0 \text{ with } \nu = 0$$

This is **Bessel's equation of order** $\nu = 0$.

General solution. For order $\nu = 0$ the two independent solutions are $J_0(z)$ and the Bessel function of the second kind, $Y_0(z)$:

$$y = c_1 J_0(z) + c_2 Y_0(z) = c_1 J_0(\sqrt{2x}) + c_2 Y_0(\sqrt{2x})$$

In series form, the first solution is

$$J_0(\sqrt{2x}) = \sum_{m=0}^{\infty} \frac{(-1)^m}{(m!)^2} \left(\frac{x}{2} \right)^m = 1 - \frac{x}{2} + \frac{x^2}{16} - \frac{x^3}{288} + \dots$$

Answer. $y = c_1 J_0(\sqrt{2x}) + c_2 Y_0(\sqrt{2x})$.

Question 5: Eigenvalues and Eigenvectors

(a) Eigenvalues and eigenvectors of a 3×3 matrix

Question 5(a). Find the eigenvalues and eigenvectors of the matrix (6 marks)

$$A = \begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix}$$

Eigenvalues. A is upper triangular, so $\det(A - \lambda I)$ is the product of the diagonal entries:

$$\det(A - \lambda I) = (3 - \lambda)(2 - \lambda)(5 - \lambda) = 0 \Rightarrow \lambda = 2, 3, 5$$

(Expanded, the characteristic equation is $\lambda^3 - 10\lambda^2 + 31\lambda - 30 = 0$.)

Eigenvectors. Solve $(A - \lambda I)x = 0$ for each eigenvalue.

$$\lambda = 2: \begin{bmatrix} 1 & 1 & 4 \\ 0 & 0 & 6 \\ 0 & 0 & 3 \end{bmatrix} x = 0 \Rightarrow x_3 = 0, x_1 + x_2 = 0 \Rightarrow x = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

$$\lambda = 3: \begin{bmatrix} 0 & 1 & 4 \\ 0 & -1 & 6 \\ 0 & 0 & 2 \end{bmatrix} x = 0 \Rightarrow x_3 = 0, x_2 = 0, x_1 \text{ free} \Rightarrow x = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\lambda = 5: \begin{bmatrix} -2 & 1 & 4 \\ 0 & -3 & 6 \\ 0 & 0 & 0 \end{bmatrix} x = 0 \Rightarrow x_2 = 2x_3, -2x_1 + 2x_3 + 4x_3 = 0 \Rightarrow x_1 = 3x_3 \Rightarrow x = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$$

Check for $\lambda = 5$: $A \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 9+2+4 \\ 4+6 \\ 5 \end{bmatrix} = \begin{bmatrix} 15 \\ 10 \\ 5 \end{bmatrix} = 5 \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} \checkmark$

Answer. $\lambda_1 = 2$ with $x_1 = k(1, -1, 0)^T$; $\lambda_2 = 3$ with $x_2 = k(1, 0, 0)^T$; $\lambda_3 = 5$ with $x_3 = k(3, 2, 1)^T$ ($k \neq 0$).

(b) Principal directions of a stretched membrane

Question 5(b). An elastic membrane in the x_1x_2 -plane with boundary circle $x_1^2 + x_2^2 = 1$ is stretched so that a point $P:(x_1, x_2)$ goes over into the point $Q:(y_1, y_2)$ given by

$$y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = A x = \begin{bmatrix} 5 & 3 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Find the principal directions, that is, the directions of the position vector x of P for which the direction of the position vector y of Q is the same or exactly opposite. (6 marks)

We need the vectors x for which $y = Ax$ is parallel to x , that is $Ax = \lambda x$. These are the eigenvectors of

$$A = \begin{bmatrix} 5 & 3 \\ 3 & 5 \end{bmatrix}$$

Eigenvalues.

$$\det(A - \lambda I) = (5 - \lambda)^2 - 9 = \lambda^2 - 10\lambda + 16 = (\lambda - 8)(\lambda - 2) = 0 \Rightarrow \lambda_1 = 8, \lambda_2 = 2$$

Eigenvectors.

For $\lambda_1 = 8$: $-3x_1 + 3x_2 = 0 \Rightarrow x_2 = x_1$, so $x^{(1)} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, at 45° to the x_1 -axis.

For $\lambda_2 = 2$: $3x_1 + 3x_2 = 0 \Rightarrow x_2 = -x_1$, so $x^{(2)} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, at 135° (or -45°) to the x_1 -axis.

Both eigenvalues are positive, so in each principal direction y points the **same** way as x . The two directions are perpendicular because A is symmetric.

Interpretation. Along the 45° direction the membrane is stretched by a factor of 8, and along the 135° direction by a factor of 2. With coordinates u_1, u_2 along these principal directions, the boundary circle $x_1^2 + x_2^2 = 1$ becomes the ellipse

$$\frac{u_1^2}{8^2} + \frac{u_2^2}{2^2} = 1$$

Answer. The principal directions are $x_2 = x_1$ (at 45° , eigenvalue 8, stretch factor 8) and $x_2 = -x_1$ (at 135° , eigenvalue 2, stretch factor 2). The circle is deformed into an ellipse with semi-axes 8 and 2 along these directions.

Question 6: Legendre's Equation and the Wave Equation

(a) Legendre's equation

Question 6(a). Solve this special Legendre equation, $(1-x^2)y'' - 2xy' + 2y = 0$, which occurs in models exhibiting spherical symmetry. (6 marks)

This is Legendre's equation $(1-x^2)y'' - 2xy' + n(n+1)y = 0$ with $n(n+1) = 2$, that is $n = 1$.

Power series. Put $y = \sum_{m=0}^{\infty} a_m x^m$. Then

$$(1-x^2) \sum_m m(m-1)a_m x^{m-2} - 2x \sum_m m a_m x^{m-1} + 2 \sum_m a_m x^m = 0$$

Collecting the coefficient of x^m :

$$(m+2)(m+1)a_{m+2} - [m(m-1) + 2m - 2]a_m = 0$$

Since $m(m-1) + 2m - 2 = m^2 + m - 2 = (m+2)(m-1)$, the recurrence relation is

$$a_{m+2} = \frac{(m+2)(m-1)}{(m+2)(m+1)} a_m = \frac{m-1}{m+1} a_m$$

Odd series ($m=1$): $a_3 = 0 \cdot a_1 = 0$, so $a_5 = a_7 = \dots = 0$ and the series stops:

$$y_1 = a_1 x = a_1 P_1(x)$$

Even series (from a_0):

$$a_2 = -a_0, a_4 = \frac{1}{3} a_2 = -\frac{a_0}{3}, a_6 = \frac{3}{5} a_4 = -\frac{a_0}{5}, a_8 = -\frac{a_0}{7}, \dots$$

$$y_2 = a_0 \left(1 - x^2 - \frac{x^4}{3} - \frac{x^6}{5} - \frac{x^8}{7} - \dots \right) = a_0 \left(1 - \frac{x}{2} \ln \frac{1+x}{1-x} \right), \forall x \in (-1, 1)$$

because $\frac{1}{2} \ln \frac{1+x}{1-x} = x + \frac{x^3}{3} + \frac{x^5}{5} + \dots$.

Check of $y_1=x$: $(1-x^2)(0)-2x(1)+2x=0$. ✓

Answer. $y=c_1x+c_2\left(1-x^2-\frac{x^4}{3}-\frac{x^6}{5}-\dots\right)=c_1P_1(x)+c_2\left(1-\frac{x}{2}\ln\frac{1+x}{1-x}\right)$ for $|x|<1$. The polynomial solution is the Legendre polynomial $P_1(x)=x$.

(b) One-dimensional wave equation

Question 6(b). Find the solution to this one-dimensional wave equation:

$\frac{\partial^2 u}{\partial x^2} = \frac{1}{16} \frac{\partial^2 u}{\partial t^2}$ for $0 < x < 2, t > 0$. The boundary conditions are $u(0, t) = u(2, t) = 0$. The initial conditions are (i) $u(x, 0) = 6 \sin \pi x - 3 \sin 4 \pi x$ and (ii) $\frac{\partial u}{\partial t}(x, 0) = 0$. (6 marks)

The equation is

$$\frac{\partial^2 u}{\partial x^2} = \frac{1}{16} \frac{\partial^2 u}{\partial t^2} \Leftrightarrow \frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}, c^2 = 16, c = 4, L = 2$$

Separation of variables. Put $u = F(x)G(t)$:

$$\frac{F''}{F} = \frac{G''}{16G} = -p^2$$

so $F'' + p^2 F = 0$ and $G'' + 16p^2 G = 0$.

Boundary conditions. $u(0, t) = u(2, t) = 0$ need $F(0) = F(2) = 0$. With $F = A \cos px + B \sin px$, $F(0) = 0$ gives $A = 0$, and $\sin 2p = 0$ gives

$$p = \frac{n\pi}{2}, F_n(x) = \sin \frac{n\pi x}{2}, n = 1, 2, 3, \dots$$

Time part. $G'' + \lambda_n^2 G_n = 0$ with $\lambda_n = \frac{cn\pi}{L} = \frac{4n\pi}{2} = 2n\pi$:

$$G_n(t) = B_n \cos 2n\pi t + B_n^i \sin 2n\pi t$$

so

$$u(x, t) = \sum_{n=1}^{\infty} \left(B_n \cos 2n\pi t + B_n^i \sin 2n\pi t \right) \sin \frac{n\pi x}{2}$$

Initial velocity. $\frac{\partial u}{\partial t}(x, 0) = \sum 2n\pi B_n^i \sin \frac{n\pi x}{2} = 0$, so $B_n^i = 0$ for every n .

Initial displacement.

$$u(x, 0) = \sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{2} = 6 \sin \pi x - 3 \sin 4\pi x$$

$\sin \pi x = \sin \frac{2\pi x}{2}$ is the $n=2$ term and $\sin 4\pi x = \sin \frac{8\pi x}{2}$ is the $n=8$ term. Comparing coefficients (no Fourier integrals are needed):

$$B_2 = 6, B_8 = -3, B_n = 0 \text{ otherwise}$$

With $\lambda_2 = 4\pi$ and $\lambda_8 = 16\pi$:

| **Answer.** $u(x, t) = 6 \sin \pi x \cos 4\pi t - 3 \sin 4\pi x \cos 16\pi t$.

Check. $u(0, t) = u(2, t) = 0$ since $\sin 2\pi = \sin 8\pi = 0$; $u(x, 0) = 6 \sin \pi x - 3 \sin 4\pi x$; $u_t(x, 0) = 0$; and each term satisfies $u_{tt} = 16u_{xx}$, for example $(4\pi)^2 = 16\pi^2$. ✓

Question 7: Numerical Differentiation

(a) Rate of population growth

Question 7(a). The table below gives the census population (in millions) of a state for the years 1961 to 2001. Find the rate of growth of the population in the year 2001. (6 marks)

Year (x)	1961	1971	1981	1991	2001
Population (y)	19.96	36.65	58.81	77.21	94.61

The year 2001 is at the **end** of the table, so we use Newton's backward difference formula with $h=10$ years.

Backward difference table.

Year x	y	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$
1961	19.96				
1971	36.65	16.69			
1981	58.81	22.16	5.47		
1991	77.21	18.40	-3.76	-9.23	
2001	94.61	17.40	-1.00	2.76	11.99

Derivative at the last point $x_n = 2001$ (where $p=0$):

$$\left(\frac{dy}{dx} \right)_{x_n} = \frac{1}{h} \left[\nabla y_n + \frac{1}{2} \nabla^2 y_n + \frac{1}{3} \nabla^3 y_n + \frac{1}{4} \nabla^4 y_n \right]$$

$$\therefore \frac{1}{10} \left[17.40 + \frac{-1.00}{2} + \frac{2.76}{3} + \frac{11.99}{4} \right] = \frac{1}{10} [17.40 - 0.50 + 0.92 + 2.9975] = \frac{20.8175}{10} = 2.0818$$

| **Answer.** In 2001 the population was growing at about **2.08 million per year**.

(b) Derivatives from tabulated data

Question 7(b). Given the following data points, determine $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ at $x=1.1$ and $x=1.5$. (6 marks)

x	1.0	1.1	1.2	1.3	1.4	1.5	1.6
$f(x)$	7.989	8.403	8.781	9.129	9.451	9.750	10.031

Here $h=0.1$. The point $x=1.1$ is near the start of the table, so we use Newton's **forward** formula there; $x=1.5$ is near the end, so we use the **backward** formula.

Difference table (row x_i lists $\Delta^k f_i$, the forward differences starting at that row).

x	$f(x)$	Δ	Δ^2	Δ^3	Δ^4	Δ^5	Δ^6
1.0	7.989	0.414	-0.036	0.006	-0.002	0.001	0.002
1.1	8.403	0.378	-0.030	0.004	-0.001	0.003	
1.2	8.781	0.348	-0.026	0.003	0.002		
1.3	9.129	0.322	-0.023	0.005			
1.4	9.451	0.299	-0.018				
1.5	9.750	0.281					
1.6	10.031						

At $x=1.1$ (forward differences, $x_0=1.1$)

From the table, $\Delta y_0=0.378$, $\Delta^2 y_0=-0.030$, $\Delta^3 y_0=0.004$, $\Delta^4 y_0=-0.001$ and $\Delta^5 y_0=0.003$

$$\left(\frac{dy}{dx}\right)_{x_0} = \frac{1}{h} \left[\Delta y_0 - \frac{1}{2} \Delta^2 y_0 + \frac{1}{3} \Delta^3 y_0 - \frac{1}{4} \Delta^4 y_0 + \frac{1}{5} \Delta^5 y_0 \right]$$

$$\therefore \frac{1}{0.1} \left[0.378 + \frac{0.030}{2} + \frac{0.004}{3} + \frac{0.001}{4} + \frac{0.003}{5} \right] = 10 [0.378 + 0.015 + 0.00133 + 0.00025 + 0.0006] = 3.952$$

$$\left(\frac{d^2y}{dx^2}\right)_{x_0} = \frac{1}{h^2} \left[\Delta^2 y_0 - \Delta^3 y_0 + \frac{11}{12} \Delta^4 y_0 - \frac{5}{6} \Delta^5 y_0 \right]$$

$$\therefore 100 \left[-0.030 - 0.004 + \frac{11}{12} (-0.001) - \frac{5}{6} (0.003) \right] = 100 [-0.030 - 0.004 - 0.00092 - 0.0025] = -3.742$$

At $x=1.5$ (backward differences, $x_n=1.5$)

From the table, $\nabla y_n=0.299$, $\nabla^2 y_n=-0.023$, $\nabla^3 y_n=0.003$, $\nabla^4 y_n=-0.001$ and $\nabla^5 y_n=0.001$.

$$\left(\frac{dy}{dx}\right)_{x_n} = \frac{1}{h} \left[\nabla y_n + \frac{1}{2} \nabla^2 y_n + \frac{1}{3} \nabla^3 y_n + \frac{1}{4} \nabla^4 y_n + \frac{1}{5} \nabla^5 y_n \right]$$

$$\therefore 10 [0.299 - 0.0115 + 0.001 - 0.00025 + 0.0002] = 2.885$$

$$\left(\frac{d^2y}{dx^2}\right)_{x_n} = \frac{1}{h^2} \left[\nabla^2 y_n + \nabla^3 y_n + \frac{11}{12} \nabla^4 y_n + \frac{5}{6} \nabla^5 y_n \right]$$

$$\approx 100[-0.023+0.003-0.00092+0.00083]=-2.008$$

Answer. At $x=1.1$: $\frac{dy}{dx} \approx 3.952$ and $\frac{d^2y}{dx^2} \approx -3.742$. At $x=1.5$: $\frac{dy}{dx} \approx 2.885$ and $\frac{d^2y}{dx^2} \approx -2.008$.

Type 2

Question 2: Differential Equations and Damped Vibration

(a) Non-homogeneous second-order ODE

Question 2(a). An engineering problem is represented by the differential equation below. Give the solution to the problem. (6 marks)

$$\frac{d^2 y}{dx^2} + \frac{dy}{dx} + y = x^2 + x e^{2x} + e^x \cos 2x$$

Complementary function. The auxiliary equation is

$$m^2 + m + 1 = 0 \Rightarrow m = \frac{-1 \pm \sqrt{1-4}}{2} = -\frac{1}{2} \pm \frac{\sqrt{3}}{2}i$$

$$y_c = e^{-x/2} \left(A \cos \frac{\sqrt{3}}{2}x + B \sin \frac{\sqrt{3}}{2}x \right)$$

Particular integral. $y_p = y_1 + y_2 + y_3$, one part for each term on the right. None of the terms appears in y_c .

For x^2 . Try $y_1 = ax^2 + bx + c$:

$$2a + (2ax + b) + (ax^2 + bx + c) = ax^2 + (2a + b)x + (2a + b + c) = x^2$$

$$a = 1, 2a + b = 0 \Rightarrow b = -2, 2a + b + c = 0 \Rightarrow c = 0 \Rightarrow y_1 = x^2 - 2x$$

For $x e^{2x}$. Put $y_2 = u e^{2x}$. Then $y_2' = (u' + 2u) e^{2x}$, $y_2'' = (u'' + 4u' + 4u) e^{2x}$ and

$$y_2'' + y_2' + y_2 = (u'' + 5u' + 7u) e^{2x} = x e^{2x}$$

Try $u = px + q$: $5p + 7(px + q) = x \Rightarrow p = 1/7, q = -5/49$, so

$$y_2 = \frac{7x - 5}{49} e^{2x}$$

For $e^x \cos 2x$. Put $y_3 = v e^x$. Then $y_3' = (v' + v) e^x$, $y_3'' = (v'' + 2v' + v) e^x$ and

$$y_3'' + y_3' + y_3 = (v'' + 3v' + 3v) e^x = e^x \cos 2x$$

Try $v = C \cos 2x + D \sin 2x$, with $v'' = -4v$:

$$(-C + 6D) \cos 2x + (-D - 6C) \sin 2x = \cos 2x \Rightarrow -C + 6D = 1, D = -6C$$

so $-37C = 1$, $C = -1/37$, $D = 6/37$, and

$$y_3 = \frac{e^x}{37} (6 \sin 2x - \cos 2x)$$

$$\boxed{\text{Answer. } y = e^{-x/2} \left(A \cos \frac{\sqrt{3}}{2}x + B \sin \frac{\sqrt{3}}{2}x \right) + x^2 - 2x + \frac{7x - 5}{49} e^{2x} + \frac{e^x}{37} (6 \sin 2x - \cos 2x)}$$

(b) Damped spring-mass system

Question 2(b). A spring with a 2000-g mass has a natural length 0.5 m, and a 25.6-newton force is required to maintain it stretched to a length of 0.7 m. If the spring is immersed in a fluid with damping constant $\beta=40$ and the mass is released from the equilibrium position with initial velocity 0.6 m/s: (i) characterize the damped spring-mass system with reason(s); (ii) find the position of the mass at any time t . (6 marks)

Data. $m=2$ kg. From Hooke's law, $k(0.7 - 0.5)=25.6$, so $k=128$ N/m. The damping constant is $\beta=40$.

Equation of motion.

$$m \frac{d^2 x}{dt^2} + \beta \frac{dx}{dt} + kx = 0 \Rightarrow 2x'' + 40x' + 128x = 0 \Rightarrow x'' + 20x' + 64x = 0$$

(i) Characterisation. The type of motion depends on the sign of $\beta^2 - 4mk$:

$$\beta^2 - 4mk = 40^2 - 4(2)(128) = 1600 - 1024 = 576 > 0$$

The auxiliary equation has two distinct real roots,

$$r^2 + 20r + 64 = (r+4)(r+16) = 0 \Rightarrow r = -4, -16$$

so the system is **overdamped**: the damping is strong enough that the mass does not oscillate, and it returns to equilibrium without passing through it more than once.

(ii) Position.

$$x(t) = c_1 e^{-4t} + c_2 e^{-16t}$$

The mass starts at equilibrium with velocity 0.6 m/s, so $x(0)=0$ and $x'(0)=0.6$.

$$c_1 + c_2 = 0, -4c_1 - 16c_2 = 0.6 \Rightarrow 12c_1 = 0.6, c_1 = 0.05, c_2 = -0.05$$

Answer. (i) Overdamped, because $\beta^2 - 4mk = 576 > 0$ (real roots -4 and -16 , no oscillation). (ii) $x(t) = 0.05(e^{-4t} - e^{-16t})$ m.

Question 3: RLC Circuit

Question 3. Find (a) the charge (6 marks) and (b) the current (6 marks) at time t in a circuit with $R=40\ \Omega$, $L=1$ H, $C=16 \times 10^{-4}$ F and $E(t)=50\cos 5t$. The initial charge and current are both 0.

The circuit equation. With $1/C=1/(16 \times 10^{-4})=625$,

$$L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{q}{C} = E(t) \Rightarrow \frac{d^2 q}{dt^2} + 40 \frac{dq}{dt} + 625q = 50\cos 5t$$

with $q(0)=0$ and $q'(0)=I(0)=0$.

Complementary function. $r^2 + 40r + 625 = 0$ gives $r = -20 \pm \sqrt{400 - 625} = -20 \pm 15i$:

$$q_c = e^{-20t} (c_1 \cos 15t + c_2 \sin 15t)$$

Particular integral. Try $q_p = A \cos 5t + B \sin 5t$, with $q_p'' = -25q_p$:

$$(625 - 25)(A \cos 5t + B \sin 5t) + 40(-5A \sin 5t + 5B \cos 5t) = 50 \cos 5t$$

$$\cos 5t: 600A + 200B = 50, \sin 5t: 600B - 200A = 0$$

The second gives $A = 3B$; then $1800B + 200B = 50$, so $B = 1/40$ and $A = 3/40$:

$$q_p = \frac{1}{40} (3 \cos 5t + \sin 5t)$$

Constants. $q(0) = c_1 + 3/40 = 0 \Rightarrow c_1 = -3/40$.

$$q'(0) = -20c_1 + 15c_2 + 5 \cdot 1/40 = 3/2 + 15c_2 + 1/8 = 0 \Rightarrow c_2 = -13/120.$$

(a) Charge.

$$q(t) = -e^{-20t} \left(\frac{3}{40} \cos 15t + \frac{13}{120} \sin 15t \right) + \frac{1}{40} (3 \cos 5t + \sin 5t)$$

(b) Current. Differentiating,

$$I(t) = \frac{dq}{dt} = e^{-20t} \left(\frac{79}{24} \sin 15t - \frac{1}{8} \cos 15t \right) + \frac{1}{8} (\cos 5t - 3 \sin 5t)$$

Check: $I(0) = -1/8 + 1/8 = 0$. ✓

Answer. (a) $q = -e^{-20t} (0.075 \cos 15t + 0.1083 \sin 15t) + 0.075 \cos 5t + 0.025 \sin 5t$ C. (b) $I = e^{-20t} (3.2917 \sin 15t - 0.125 \cos 15t) + 0.125 \cos 5t - 0.375 \sin 5t$ A. The e^{-20t} terms are the transient; the steady-state current is $1/8 (\cos 5t - 3 \sin 5t)$.

Question 4: Series Solutions and Bessel's Equation

(a) Power series solution

Question 4(a). Find the power series solution of $2(y'' + 2y) = 0$. (6 marks)

Dividing by 2, the equation is $y'' + 2y = 0$. Assume $y = \sum_{n=0}^{\infty} a_n x^n$, so

$y'' = \sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2}x^n$. Substituting:

$$\sum_{n=0}^{\infty} [(n+2)(n+1)a_{n+2} + 2a_n]x^n = 0 \Rightarrow a_{n+2} = -\frac{2a_n}{(n+2)(n+1)}$$

Even coefficients:

$$a_2 = -\frac{2a_0}{2!}, a_4 = -\frac{2a_2}{4 \cdot 3} = \frac{2^2 a_0}{4!}, a_6 = -\frac{2^3 a_0}{6!}, \dots$$

Odd coefficients:

$$a_3 = -\frac{2a_1}{3!}, a_5 = \frac{2^2 a_1}{5!}, a_7 = -\frac{2^3 a_1}{7!}, \dots$$

Hence

$$y = a_0 \left(1 - \frac{2x^2}{2!} + \frac{4x^4}{4!} - \frac{8x^6}{6!} + \dots \right) + a_1 \left(x - \frac{2x^3}{3!} + \frac{4x^5}{5!} - \frac{8x^7}{7!} + \dots \right)$$

Since $2^k x^{2k} = (\sqrt{2}x)^{2k}$, the first series is $\cos \sqrt{2}x$ and the second is $\frac{1}{\sqrt{2}} \sin \sqrt{2}x$.

$$\left| \text{Answer. } y = a_0 \left(1 - x^2 + \frac{x^4}{6} - \dots \right) + a_1 \left(x - \frac{x^3}{3} + \frac{x^5}{30} - \dots \right) = a_0 \cos \sqrt{2}x + \frac{a_1}{\sqrt{2}} \sin \sqrt{2}x \right.$$

(b) Bessel's equation

Question 4(b). Find the general solution, in terms of Bessel functions, of the equation $x y'' + y' + \left(\frac{2}{x}\right) y = 0$. (6 marks)

The fraction in the equation reduces to $1/2$, so the equation is

$$x y'' + y' + 1/2 y = 0$$

Change of variable. Equations of the form $x y'' + y' + \lambda y = 0$ become Bessel's equation under $z = 2\sqrt{\lambda x}$. Here $\lambda = 1/2$, so put

$$z = 2\sqrt{x/2} = \sqrt{2x}, z^2 = 2x, \frac{dz}{dx} = \frac{1}{z}$$

Then

$$\frac{dy}{dx} = \frac{1}{z} \frac{dy}{dz}, \frac{d^2 y}{dx^2} = \frac{1}{z} \frac{d}{dz} \left(\frac{1}{z} \frac{dy}{dz} \right) = \frac{1}{z^2} \frac{d^2 y}{dz^2} - \frac{1}{z^3} \frac{dy}{dz}$$

With $x = z^2/2$:

$$x \frac{d^2 y}{dx^2} + \frac{dy}{dx} + \frac{1}{2} y = \frac{1}{2} \frac{d^2 y}{dz^2} - \frac{1}{2z} \frac{dy}{dz} + \frac{1}{z} \frac{dy}{dz} + \frac{1}{2} y = \frac{1}{2} \left(\frac{d^2 y}{dz^2} + \frac{1}{z} \frac{dy}{dz} + y \right) = 0$$

Multiplying by $2z^2$:

$$z^2 \frac{d^2 y}{dz^2} + z \frac{dy}{dz} + (z^2 - \nu^2) y = 0 \text{ with } \nu = 0$$

This is **Bessel's equation of order** $\nu = 0$.

General solution. For order $\nu = 0$ the two independent solutions are $J_0(z)$ and the Bessel function of the second kind, $Y_0(z)$:

$$y = c_1 J_0(z) + c_2 Y_0(z) = c_1 J_0(\sqrt{2x}) + c_2 Y_0(\sqrt{2x})$$

In series form, the first solution is

$$J_0(\sqrt{2x}) = \sum_{m=0}^{\infty} \frac{(-1)^m}{(m!)^2} \left(\frac{x}{2}\right)^m = 1 - \frac{x}{2} + \frac{x^2}{16} - \frac{x^3}{288} + \dots$$

Answer. $y = c_1 J_0(\sqrt{2x}) + c_2 Y_0(\sqrt{2x})$.

Question 5: Eigenvalues and Eigenvectors

(a) Eigenvalues and eigenvectors of a 3×3 matrix

Question 5(a). Find the eigenvalues and eigenvectors of the matrix (6 marks)

$$A = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}$$

Characteristic equation. Expanding along the first row,

$$\det(A - \lambda I) = (1 - \lambda)[(2 - \lambda)(1 - \lambda) - 1] + 1 \cdot [-(1 - \lambda)] = (1 - \lambda)[(2 - \lambda)(1 - \lambda) - 2]$$

$$\therefore (1 - \lambda)(\lambda^2 - 3\lambda) = -\lambda(\lambda - 1)(\lambda - 3) = 0$$

so $\lambda = 0, 1, 3$. (Expanded: $\lambda^3 - 4\lambda^2 + 3\lambda = 0$.)

Eigenvectors.

$\lambda = 0$: $x_1 - x_2 = 0$ and $-x_2 + x_3 = 0 \Rightarrow x = (1, 1, 1)^T$

$\lambda = 1$: $\begin{bmatrix} 0 & -1 & 0 \\ -1 & 1 & -1 \\ 0 & -1 & 0 \end{bmatrix} x = 0 \Rightarrow x_2 = 0, x_1 = -x_3 \Rightarrow x = (1, 0, -1)^T$

$\lambda = 3$: $\begin{bmatrix} -2 & -1 & 0 \\ -1 & -1 & -1 \\ 0 & -1 & -2 \end{bmatrix} x = 0 \Rightarrow x_2 = -2x_1, x_2 = -2x_3 \Rightarrow x = (1, -2, 1)^T$

Check for $\lambda = 3$: $A(1, -2, 1)^T = (1+2, -1-4-1, 2+1)^T = (3, -6, 3)^T = 3(1, -2, 1)^T \checkmark$. The matrix is symmetric, so the three eigenvectors are mutually orthogonal.

Answer. $\lambda_1 = 0$ with $x_1 = k(1, 1, 1)^T$; $\lambda_2 = 1$ with $x_2 = k(1, 0, -1)^T$; $\lambda_3 = 3$ with $x_3 = k(1, -2, 1)^T$ ($k \neq 0$).

(b) Principal directions of a stretched membrane

Question 5(b). An elastic membrane in the x_1x_2 -plane with boundary circle $x_1^2 + x_2^2 = 1$ is stretched so that a point $P:(x_1, x_2)$ goes over into the point $Q:(y_1, y_2)$ given by

$$y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = A x = \begin{bmatrix} 5 & 3 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Find the principal directions, that is, the directions of the position vector x of P for which the direction of the position vector y of Q is the same or exactly opposite. (6 marks)

We need the vectors x for which $y = Ax$ is parallel to x , that is $Ax = \lambda x$. These are the eigenvectors of

$$A = \begin{bmatrix} 5 & 3 \\ 3 & 5 \end{bmatrix}$$

Eigenvalues.

$$\det(A - \lambda I) = (5 - \lambda)^2 - 9 = \lambda^2 - 10\lambda + 16 = (\lambda - 8)(\lambda - 2) = 0 \Rightarrow \lambda_1 = 8, \lambda_2 = 2$$

Eigenvectors.

For $\lambda_1 = 8$: $-3x_1 + 3x_2 = 0 \Rightarrow x_2 = x_1$, so $x^{(1)} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, at 45° to the x_1 -axis.

For $\lambda_2 = 2$: $3x_1 + 3x_2 = 0 \Rightarrow x_2 = -x_1$, so $x^{(2)} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, at 135° (or -45°) to the x_1 -axis.

Both eigenvalues are positive, so in each principal direction y points the **same** way as x . The two directions are perpendicular because A is symmetric.

Interpretation. Along the 45° direction the membrane is stretched by a factor of 8, and along the 135° direction by a factor of 2. With coordinates u_1, u_2 along these principal directions, the boundary circle $x_1^2 + x_2^2 = 1$ becomes the ellipse

$$\frac{u_1^2}{8^2} + \frac{u_2^2}{2^2} = 1$$

Answer. The principal directions are $x_2 = x_1$ (at 45° , eigenvalue 8, stretch factor 8) and $x_2 = -x_1$ (at 135° , eigenvalue 2, stretch factor 2). The circle is deformed into an ellipse with semi-axes 8 and 2 along these directions.

Question 6: Numerical Differentiation

(a) Rate of population growth

Question 6(a). The table below gives the census population (in millions) of a state for the years 1961 to 2001. Find the rate of growth of the population in the year 2001. (6 marks)

Year (x)	1961	1971	1981	1991	2001
Population (y)	19.96	36.65	58.81	77.21	94.61

The year 2001 is at the **end** of the table, so we use Newton's backward difference formula with $h = 10$ years.

Backward difference table.

Year x	y	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$
1961	19.96				
1971	36.65	16.69			
1981	58.81	22.16	5.47		
1991	77.21	18.40	-3.76	-9.23	
2001	94.61	17.40	-1.00	2.76	11.99

Derivative at the last point $x_n=2001$ (where $p=0$):

$$\left(\frac{dy}{dx}\right)_{x_n} = \frac{1}{h} \left[\nabla y_n + \frac{1}{2} \nabla^2 y_n + \frac{1}{3} \nabla^3 y_n + \frac{1}{4} \nabla^4 y_n \right]$$

$$\hat{=} \frac{1}{10} \left[17.40 + \frac{-1.00}{2} + \frac{2.76}{3} + \frac{11.99}{4} \right] = \frac{1}{10} [17.40 - 0.50 + 0.92 + 2.9975] = \frac{20.8175}{10} = 2.0818$$

| **Answer.** In 2001 the population was growing at about **2.08 million per year**.

(b) Derivatives from tabulated data

Question 6(b). Given the following data points, determine $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ at $x=1.1$ and $x=1.5$. (6 marks)

x	1.0	1.1	1.2	1.3	1.4	1.5	1.6
$f(x)$	7.989	8.403	8.781	9.129	9.451	9.750	10.031

Here $h=0.1$. The point $x=1.1$ is near the start of the table, so we use Newton's **forward** formula there; $x=1.5$ is near the end, so we use the **backward** formula.

Difference table (row x_i lists $\Delta^k f_i$, the forward differences starting at that row).

x	$f(x)$	Δ	Δ^2	Δ^3	Δ^4	Δ^5	Δ^6
1.0	7.989	0.414	-0.036	0.006	-0.002	0.001	0.002
1.1	8.403	0.378	-0.030	0.004	-0.001	0.003	
1.2	8.781	0.348	-0.026	0.003	0.002		
1.3	9.129	0.322	-0.023	0.005			
1.4	9.451	0.299	-0.018				
1.5	9.750	0.281					
1.6	10.031						

At $x=1.1$ (forward differences, $x_0=1.1$)

From the table, $\Delta y_0=0.378$, $\Delta^2 y_0=-0.030$, $\Delta^3 y_0=0.004$, $\Delta^4 y_0=-0.001$ and $\Delta^5 y_0=0.003$

$$\left(\frac{dy}{dx}\right)_{x_0} = \frac{1}{h} \left[\Delta y_0 - \frac{1}{2} \Delta^2 y_0 + \frac{1}{3} \Delta^3 y_0 - \frac{1}{4} \Delta^4 y_0 + \frac{1}{5} \Delta^5 y_0 \right]$$

$$\therefore \frac{1}{0.1} \left[0.378 + \frac{0.030}{2} + \frac{0.004}{3} + \frac{0.001}{4} + \frac{0.003}{5} \right] = 10 [0.378 + 0.015 + 0.00133 + 0.00025 + 0.0006] = 3.952$$

$$\left(\frac{d^2y}{dx^2}\right)_{x_0} = \frac{1}{h^2} \left[\Delta^2 y_0 - \Delta^3 y_0 + \frac{11}{12} \Delta^4 y_0 - \frac{5}{6} \Delta^5 y_0 \right]$$

$$\therefore 100 \left[-0.030 - 0.004 + \frac{11}{12} (-0.001) - \frac{5}{6} (0.003) \right] = 100 [-0.030 - 0.004 - 0.00092 - 0.0025] = -3.742$$

At $x=1.5$ (backward differences, $x_n=1.5$)

From the table, $\nabla y_n=0.299$, $\nabla^2 y_n=-0.023$, $\nabla^3 y_n=0.003$, $\nabla^4 y_n=-0.001$ and $\nabla^5 y_n=0.001$.

$$\left(\frac{dy}{dx}\right)_{x_n} = \frac{1}{h} \left[\nabla y_n + \frac{1}{2} \nabla^2 y_n + \frac{1}{3} \nabla^3 y_n + \frac{1}{4} \nabla^4 y_n + \frac{1}{5} \nabla^5 y_n \right]$$

$$\therefore 10 [0.299 - 0.0115 + 0.001 - 0.00025 + 0.0002] = 2.885$$

$$\left(\frac{d^2y}{dx^2}\right)_{x_n} = \frac{1}{h^2} \left[\nabla^2 y_n + \nabla^3 y_n + \frac{11}{12} \nabla^4 y_n + \frac{5}{6} \nabla^5 y_n \right]$$

$$\therefore 100 [-0.023 + 0.003 - 0.00092 + 0.00083] = -2.008$$

Answer. At $x=1.1$: $\frac{dy}{dx} \approx 3.952$ and $\frac{d^2y}{dx^2} \approx -3.742$. At $x=1.5$: $\frac{dy}{dx} \approx 2.885$ and $\frac{d^2y}{dx^2} \approx -2.008$.

Question 7: Legendre's Equation and the Wave Equation

(a) Legendre's equation

Question 7(a). Solve this special Legendre equation, $(1-x^2)y'' - 2xy' + 2y = 0$, which occurs in models exhibiting spherical symmetry. (6 marks)

This is Legendre's equation $(1-x^2)y'' - 2xy' + n(n+1)y = 0$ with $n(n+1)=2$, that is $n=1$

Power series. Put $y = \sum_{m=0}^{\infty} a_m x^m$. Then

$$(1-x^2) \sum_m m(m-1) a_m x^{m-2} - 2x \sum_m m a_m x^{m-1} + 2 \sum_m a_m x^m = 0$$

Collecting the coefficient of x^m :

$$(m+2)(m+1)a_{m+2} - [m(m-1) + 2m - 2]a_m = 0$$

Since $m(m-1) + 2m - 2 = m^2 + m - 2 = (m+2)(m-1)$, the recurrence relation is

$$a_{m+2} = \frac{(m+2)(m-1)}{(m+2)(m+1)} a_m = \frac{m-1}{m+1} a_m$$

Odd series ($m=1$): $a_3=0 \cdot a_1=0$, so $a_5=a_7=\dots=0$ and the series stops:

$$y_1 = a_1 x = a_1 P_1(x)$$

Even series (from a_0):

$$a_2 = -a_0, a_4 = \frac{1}{3}a_2 = -\frac{a_0}{3}, a_6 = \frac{3}{5}a_4 = -\frac{a_0}{5}, a_8 = -\frac{a_0}{7}, \dots$$

$$y_2 = a_0 \left(1 - x^2 - \frac{x^4}{3} - \frac{x^6}{5} - \frac{x^8}{7} - \dots \right) = a_0 \left(1 - \frac{x}{2} \ln \frac{1+x}{1-x} \right), \forall x \in (-1, 1)$$

because $\frac{1}{2} \ln \frac{1+x}{1-x} = x + \frac{x^3}{3} + \frac{x^5}{5} + \dots$.

Check of $y_1 = x$: $(1-x^2)(0) - 2x(1) + 2x = 0$. ✓

Answer. $y = c_1 x + c_2 \left(1 - x^2 - \frac{x^4}{3} - \frac{x^6}{5} - \dots \right) = c_1 P_1(x) + c_2 \left(1 - \frac{x}{2} \ln \frac{1+x}{1-x} \right)$ for $|x| < 1$. The polynomial solution is the Legendre polynomial $P_1(x) = x$.

(b) One-dimensional wave equation

Question 7(b). Find the solution to this one-dimensional wave equation:

$\frac{\partial^2 u}{\partial x^2} = \frac{1}{16} \frac{\partial^2 u}{\partial t^2}$ for $0 < x < 2, t > 0$. The boundary conditions are $u(0, t) = u(2, t) = 0$. The initial conditions are (i) $u(x, 0) = 6 \sin \pi x - 3 \sin 4\pi x$ and (ii) $\frac{\partial u}{\partial t}(x, 0) = 0$. (6 marks)

The equation is

$$\frac{\partial^2 u}{\partial x^2} = \frac{1}{16} \frac{\partial^2 u}{\partial t^2} \Leftrightarrow \frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}, c^2 = 16, c = 4, L = 2$$

Separation of variables. Put $u = F(x)G(t)$:

$$\frac{F''}{F} = \frac{G''}{16G} = -p^2$$

so $F'' + p^2 F = 0$ and $G'' + 16p^2 G = 0$.

Boundary conditions. $u(0, t) = u(2, t) = 0$ need $F(0) = F(2) = 0$. With $F = A \cos px + B \sin px$, $F(0) = 0$ gives $A = 0$, and $\sin 2p = 0$ gives

$$p = \frac{n\pi}{2}, F_n(x) = \sin \frac{n\pi x}{2}, n=1, 2, 3, \dots$$

Time part. $G_n'' + \lambda_n^2 G_n = 0$ with $\lambda_n = \frac{cn\pi}{L} = \frac{4n\pi}{2} = 2n\pi$:

$$G_n(t) = B_n \cos 2n\pi t + B_n^i \sin 2n\pi t$$

so

$$u(x, t) = \sum_{n=1}^{\infty} (B_n \cos 2n\pi t + B_n^i \sin 2n\pi t) \sin \frac{n\pi x}{2}$$

Initial velocity. $\frac{\partial u}{\partial t}(x, 0) = \sum 2n\pi B_n^i \sin \frac{n\pi x}{2} = 0$, so $B_n^i = 0$ for every n .

Initial displacement.

$$u(x, 0) = \sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{2} = 6 \sin \pi x - 3 \sin 4\pi x$$

$\sin \pi x = \sin \frac{2\pi x}{2}$ is the $n=2$ term and $\sin 4\pi x = \sin \frac{8\pi x}{2}$ is the $n=8$ term. Comparing coefficients (no Fourier integrals are needed):

$$B_2 = 6, B_8 = -3, B_n = 0 \text{ otherwise}$$

With $\lambda_2 = 4\pi$ and $\lambda_8 = 16\pi$:

| **Answer.** $u(x, t) = 6 \sin \pi x \cos 4\pi t - 3 \sin 4\pi x \cos 16\pi t$.

Check. $u(0, t) = u(2, t) = 0$ since $\sin 2\pi = \sin 8\pi = 0$; $u(x, 0) = 6 \sin \pi x - 3 \sin 4\pi x$; $u_t(x, 0) = 0$; and each term satisfies $u_{tt} = 16u_{xx}$, for example $(4\pi)^2 = 16\pi^2$. ✓

Type 3

Question 2: Numerical Differentiation

(a) Rate of population growth

Question 2(a). The table below gives the census population (in millions) of a state for the years 1961 to 2001. Find the rate of growth of the population in the year 2001. (6 marks)

Year (x)	1961	1971	1981	1991	2001
Population (y)	19.96	36.65	58.81	77.21	94.61

The year 2001 is at the **end** of the table, so we use Newton's backward difference formula with $h=10$ years.

Backward difference table.

Year x	y	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$
1961	19.96				
1971	36.65	16.69			
1981	58.81	22.16	5.47		
1991	77.21	18.40	-3.76	-9.23	
2001	94.61	17.40	-1.00	2.76	11.99

Derivative at the last point $x_n=2001$ (where $p=0$):

$$\left(\frac{dy}{dx}\right)_{x_n} = \frac{1}{h} \left[\nabla y_n + \frac{1}{2} \nabla^2 y_n + \frac{1}{3} \nabla^3 y_n + \frac{1}{4} \nabla^4 y_n \right]$$

$$\therefore \frac{1}{10} \left[17.40 + \frac{-1.00}{2} + \frac{2.76}{3} + \frac{11.99}{4} \right] = \frac{1}{10} [17.40 - 0.50 + 0.92 + 2.9975] = \frac{20.8175}{10} = 2.0818$$

| **Answer.** In 2001 the population was growing at about **2.08 million per year**.

(b) Derivatives from tabulated data

Question 2(b). Given the following data points, determine $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ at $x=1.1$ and $x=1.5$. (6 marks)

x	1.0	1.1	1.2	1.3	1.4	1.5	1.6
$f(x)$	7.989	8.403	8.781	9.129	9.451	9.750	10.031

Here $h=0.1$. The point $x=1.1$ is near the start of the table, so we use Newton's **forward** formula there; $x=1.5$ is near the end, so we use the **backward** formula.

Difference table (row x_i lists $\Delta^k f_i$, the forward differences starting at that row).

x	$f(x)$	Δ	Δ^2	Δ^3	Δ^4	Δ^5	Δ^6
1.0	7.989	0.414	-0.036	0.006	-0.002	0.001	0.002
1.1	8.403	0.378	-0.030	0.004	-0.001	0.003	
1.2	8.781	0.348	-0.026	0.003	0.002		
1.3	9.129	0.322	-0.023	0.005			
1.4	9.451	0.299	-0.018				
1.5	9.750	0.281					
1.6	10.031						

At $x=1.1$ (forward differences, $x_0=1.1$)

From the table, $\Delta y_0=0.378$, $\Delta^2 y_0=-0.030$, $\Delta^3 y_0=0.004$, $\Delta^4 y_0=-0.001$ and $\Delta^5 y_0=0.003$

$$\left(\frac{dy}{dx}\right)_{x_0} = \frac{1}{h} \left[\Delta y_0 - \frac{1}{2} \Delta^2 y_0 + \frac{1}{3} \Delta^3 y_0 - \frac{1}{4} \Delta^4 y_0 + \frac{1}{5} \Delta^5 y_0 \right]$$

$$\therefore \frac{1}{0.1} \left[0.378 + \frac{0.030}{2} + \frac{0.004}{3} + \frac{0.001}{4} + \frac{0.003}{5} \right] = 10 [0.378 + 0.015 + 0.00133 + 0.00025 + 0.0006] = 3.952$$

$$\left(\frac{d^2y}{dx^2}\right)_{x_0} = \frac{1}{h^2} \left[\Delta^2 y_0 - \Delta^3 y_0 + \frac{11}{12} \Delta^4 y_0 - \frac{5}{6} \Delta^5 y_0 \right]$$

$$\therefore 100 \left[-0.030 - 0.004 + \frac{11}{12} (-0.001) - \frac{5}{6} (0.003) \right] = 100 [-0.030 - 0.004 - 0.00092 - 0.0025] = -3.742$$

At $x=1.5$ (backward differences, $x_n=1.5$)

From the table, $\nabla y_n=0.299$, $\nabla^2 y_n=-0.023$, $\nabla^3 y_n=0.003$, $\nabla^4 y_n=-0.001$ and $\nabla^5 y_n=0.001$.

$$\left(\frac{dy}{dx}\right)_{x_n} = \frac{1}{h} \left[\nabla y_n + \frac{1}{2} \nabla^2 y_n + \frac{1}{3} \nabla^3 y_n + \frac{1}{4} \nabla^4 y_n + \frac{1}{5} \nabla^5 y_n \right]$$

$$\therefore 10 [0.299 - 0.0115 + 0.001 - 0.00025 + 0.0002] = 2.885$$

$$\left(\frac{d^2y}{dx^2}\right)_{x_n} = \frac{1}{h^2} \left[\nabla^2 y_n + \nabla^3 y_n + \frac{11}{12} \nabla^4 y_n + \frac{5}{6} \nabla^5 y_n \right]$$

$$\therefore 100 [-0.023 + 0.003 - 0.00092 + 0.00083] = -2.008$$

Answer. At $x=1.1$: $\frac{dy}{dx} \approx 3.952$ and $\frac{d^2y}{dx^2} \approx -3.742$. At $x=1.5$: $\frac{dy}{dx} \approx 2.885$ and $\frac{d^2y}{dx^2} \approx -2.008$.

Question 3: RLC Circuit

Question 3. Consider a series RLC circuit consisting of a $20\ \Omega$ resistor, a $1\ \text{H}$ inductor and a $2000\ \mu\text{F}$ capacitor. The circuit starts with zero initial charge and zero initial current. (a) Determine the expression for the charge on the capacitor at time t (i) when the circuit is connected to a constant $12\ \text{V}$ DC source, and (ii) when the circuit is driven by an AC source with voltage $E(t)=12\sin 10t$. (6 marks) (b) For each case in part (a), also find the corresponding current in the circuit as a function of time. (6 marks)

The circuit equation. By Kirchhoff's voltage law, with $I=dq/dt$,

$$L \frac{d^2q}{dt^2} + R \frac{dq}{dt} + \frac{q}{C} = E(t)$$

Here $R=20\ \Omega$, $L=1\ \text{H}$ and $C=2000\ \mu\text{F}=2 \times 10^{-3}\ \text{F}$, so $1/C=500$:

$$\frac{d^2q}{dt^2} + 20 \frac{dq}{dt} + 500q = E(t), q(0)=0, I(0)=q'(0)=0$$

Complementary function. $r^2+20r+500=0$ gives $r=-10 \pm \sqrt{100-500}=-10 \pm 20i$:

$$q_c = e^{-10t}(c_1 \cos 20t + c_2 \sin 20t)$$

Case (i): 12 V battery, $E=12$

Particular integral. Try a constant: $500q_p=12$, so $q_p=\frac{12}{500}=0.024$.

$$q = e^{-10t}(c_1 \cos 20t + c_2 \sin 20t) + 0.024$$

$$q(0)=0: c_1+0.024=0 \Rightarrow c_1=-0.024.$$

$$q'(0)=0: -10c_1+20c_2=0 \Rightarrow c_2=1/2c_1=-0.012.$$

$$q(t)=0.024 - e^{-10t}(0.024 \cos 20t + 0.012 \sin 20t) = \frac{3}{250} [2 - e^{-10t}(2 \cos 20t + \sin 20t)]$$

Current. Differentiating,

$$I = \frac{dq}{dt} = 10e^{-10t}(0.024 \cos 20t + 0.012 \sin 20t) - e^{-10t}(-0.48 \sin 20t + 0.24 \cos 20t) = 0.6e^{-10t} \sin 20t$$

Case (ii): generator, $E=12\sin 10t$

Particular integral. Try $q_p=A \cos 10t+B \sin 10t$, so $q_p''=-100q_p$:

$$(500-100)(A \cos 10t+B \sin 10t)+20(-10A \sin 10t+10B \cos 10t)=12 \sin 10t$$

$$\cos 10t: 400A + 200B = 0, \sin 10t: 400B - 200A = 12$$

From the first, $B = -2A$; then $-800A - 200A = 12$, so $A = -0.012$ and $B = 0.024$:

$$q_p = -0.012 \cos 10t + 0.024 \sin 10t$$

Constants. $q(0) = c_1 - 0.012 = 0 \Rightarrow c_1 = 0.012$.

$$q'(0) = -10c_1 + 20c_2 + 10(0.024) = -0.12 + 20c_2 + 0.24 = 0 \Rightarrow c_2 = -0.006.$$

$$q(t) = e^{-10t}(0.012 \cos 20t - 0.006 \sin 20t) - 0.012 \cos 10t + 0.024 \sin 10t$$

Current.

$$I = \frac{dq}{dt} = -e^{-10t}(0.24 \cos 20t + 0.18 \sin 20t) + 0.24 \cos 10t + 0.12 \sin 10t$$

Check: $I(0) = -0.24 + 0.24 = 0$. ✓ The e^{-10t} terms are the transient, which dies away; the steady-state current is $0.24 \cos 10t + 0.12 \sin 10t$.

Answer. (i) $q = 0.024 - e^{-10t}(0.024 \cos 20t + 0.012 \sin 20t)$ C and $I = 0.6 e^{-10t} \sin 20t$ A.
(ii) $q = e^{-10t}(0.012 \cos 20t - 0.006 \sin 20t) - 0.012 \cos 10t + 0.024 \sin 10t$ C and
 $I = -e^{-10t}(0.24 \cos 20t + 0.18 \sin 20t) + 0.24 \cos 10t + 0.12 \sin 10t$ A.

Question 4: Series Solutions and Bessel's Equation

(a) Power series solution

Question 4(a). Find the power series solution of $3(y'' + y) = 0$. (6 marks)

Dividing by the non-zero constant 3 leaves $y'' + y = 0$.

Assume a power series solution about $x=0$:

$$y = \sum_{n=0}^{\infty} a_n x^n, y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} = \sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n$$

Substituting into $y'' + y = 0$:

$$\sum_{n=0}^{\infty} [(n+2)(n+1) a_{n+2} + a_n] x^n = 0$$

Every coefficient must vanish, which gives the **recurrence relation**

$$a_{n+2} = -\frac{a_n}{(n+2)(n+1)}, n=0, 1, 2, \dots$$

Even coefficients (in terms of a_0):

$$a_2 = -\frac{a_0}{2!}, a_4 = -\frac{a_2}{4 \cdot 3} = \frac{a_0}{4!}, a_6 = -\frac{a_4}{6 \cdot 5} = -\frac{a_0}{6!}, \dots$$

Odd coefficients (in terms of a_1):

$$a_3 = -\frac{a_1}{3!}, a_5 = -\frac{a_3}{5 \cdot 4} = \frac{a_1}{5!}, a_7 = -\frac{a_1}{7!}, \dots$$

Hence

$$y = a_0 \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots \right) + a_1 \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots \right)$$

The two series are the Maclaurin series of $\cos x$ and $\sin x$, and both converge for all x .

$$\boxed{\text{Answer. } y = a_0 \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots \right) + a_1 \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots \right) = a_0 \cos x + a_1 \sin x, \text{ where } a_0 \text{ and } a_1 \text{ are arbitrary constants.}}$$

(b) Bessel's equation

Question 4(b). Find the general solution, in terms of Bessel functions, of the equation $x y'' + y' + \left(\frac{3}{x}\right) y = 0$. (6 marks)

The fraction in the equation reduces to $1/2$, so the equation is

$$x y'' + y' + 1/2 y = 0$$

Change of variable. Equations of the form $x y'' + y' + \lambda y = 0$ become Bessel's equation under $z = 2\sqrt{\lambda x}$. Here $\lambda = 1/2$, so put

$$z = 2\sqrt{x/2} = \sqrt{2x}, z^2 = 2x, \frac{dz}{dx} = \frac{1}{z}$$

Then

$$\frac{dy}{dx} = \frac{1}{z} \frac{dy}{dz}, \frac{d^2y}{dx^2} = \frac{1}{z} \frac{d}{dz} \left(\frac{1}{z} \frac{dy}{dz} \right) = \frac{1}{z^2} \frac{d^2y}{dz^2} - \frac{1}{z^3} \frac{dy}{dz}$$

With $x = z^2/2$:

$$x \frac{d^2y}{dx^2} + \frac{dy}{dx} + \frac{1}{2} y = \frac{1}{2} \frac{d^2y}{dz^2} - \frac{1}{2z} \frac{dy}{dz} + \frac{1}{z} \frac{dy}{dz} + \frac{1}{2} y = \frac{1}{2} \left(\frac{d^2y}{dz^2} + \frac{1}{z} \frac{dy}{dz} + y \right) = 0$$

Multiplying by $2z^2$:

$$z^2 \frac{d^2y}{dz^2} + z \frac{dy}{dz} + (z^2 - \nu^2) y = 0 \text{ with } \nu = 0$$

This is **Bessel's equation of order** $\nu = 0$.

General solution. For order $\nu = 0$ the two independent solutions are $J_0(z)$ and the Bessel function of the second kind, $Y_0(z)$:

$$y = c_1 J_0(z) + c_2 Y_0(z) = c_1 J_0(\sqrt{2x}) + c_2 Y_0(\sqrt{2x})$$

In series form, the first solution is

$$J_0(\sqrt{2x}) = \sum_{m=0}^{\infty} \frac{(-1)^m}{(m!)^2} \left(\frac{x}{2}\right)^m = 1 - \frac{x}{2} + \frac{x^2}{16} - \frac{x^3}{288} + \dots$$

Answer. $y = c_1 J_0(\sqrt{2x}) + c_2 Y_0(\sqrt{2x})$.

Question 5: Legendre's Equation and the Wave Equation

(a) Legendre's equation

Question 5(a). Solve this special Legendre equation, $(1-x^2)y'' - 2xy' + 2y = 0$, which occurs in models exhibiting spherical symmetry. (6 marks)

This is Legendre's equation $(1-x^2)y'' - 2xy' + n(n+1)y = 0$ with $n(n+1) = 2$, that is $n = 1$.

Power series. Put $y = \sum_{m=0}^{\infty} a_m x^m$. Then

$$(1-x^2) \sum_m m(m-1)a_m x^{m-2} - 2x \sum_m m a_m x^{m-1} + 2 \sum_m a_m x^m = 0$$

Collecting the coefficient of x^m :

$$(m+2)(m+1)a_{m+2} - [m(m-1) + 2m - 2]a_m = 0$$

Since $m(m-1) + 2m - 2 = m^2 + m - 2 = (m+2)(m-1)$, the recurrence relation is

$$a_{m+2} = \frac{(m+2)(m-1)}{(m+2)(m+1)} a_m = \frac{m-1}{m+1} a_m$$

Odd series ($m=1$): $a_3 = 0 \cdot a_1 = 0$, so $a_5 = a_7 = \dots = 0$ and the series stops:

$$y_1 = a_1 x = a_1 P_1(x)$$

Even series (from a_0):

$$a_2 = -a_0, a_4 = \frac{1}{3}a_2 = -\frac{a_0}{3}, a_6 = \frac{3}{5}a_4 = -\frac{a_0}{5}, a_8 = -\frac{a_0}{7}, \dots$$

$$y_2 = a_0 \left(1 - x^2 - \frac{x^4}{3} - \frac{x^6}{5} - \frac{x^8}{7} - \dots \right) = a_0 \left(1 - \frac{x}{2} \ln \frac{1+x}{1-x} \right), \forall x \in (-1, 1)$$

because $\frac{1}{2} \ln \frac{1+x}{1-x} = x + \frac{x^3}{3} + \frac{x^5}{5} + \dots$.

Check of $y_1 = x$: $(1-x^2)(0) - 2x(1) + 2x = 0$. ✓

Answer. $y = c_1 x + c_2 \left(1 - x^2 - \frac{x^4}{3} - \frac{x^6}{5} - \dots \right) = c_1 P_1(x) + c_2 \left(1 - \frac{x}{2} \ln \frac{1+x}{1-x} \right)$ for $|x| < 1$. The polynomial solution is the Legendre polynomial $P_1(x) = x$.

(b) One-dimensional wave equation

Question 5(b). Find the solution to this one-dimensional wave equation:

$\frac{\partial^2 u}{\partial x^2} = \frac{1}{16} \frac{\partial^2 u}{\partial t^2}$ for $0 < x < 2, t > 0$. The boundary conditions are $u(0, t) = u(2, t) = 0$. The initial conditions are (i) $u(x, 0) = 6 \sin \pi x - 3 \sin 4 \pi x$ and (ii) $\frac{\partial u}{\partial t}(x, 0) = 0$. (6 marks)

The equation is

$$\frac{\partial^2 u}{\partial x^2} = \frac{1}{16} \frac{\partial^2 u}{\partial t^2} \Leftrightarrow \frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}, c^2 = 16, c = 4, L = 2$$

Separation of variables. Put $u = F(x)G(t)$:

$$\frac{F''}{F} = \frac{G''}{16G} = -p^2$$

so $F'' + p^2 F = 0$ and $G'' + 16p^2 G = 0$.

Boundary conditions. $u(0, t) = u(2, t) = 0$ need $F(0) = F(2) = 0$. With $F = A \cos px + B \sin px$, $F(0) = 0$ gives $A = 0$, and $\sin 2p = 0$ gives

$$p = \frac{n\pi}{2}, F_n(x) = \sin \frac{n\pi x}{2}, n = 1, 2, 3, \dots$$

Time part. $G'' + \lambda_n^2 G_n = 0$ with $\lambda_n = \frac{cn\pi}{L} = \frac{4n\pi}{2} = 2n\pi$:

$$G_n(t) = B_n \cos 2n\pi t + B_n^i \sin 2n\pi t$$

so

$$u(x, t) = \sum_{n=1}^{\infty} (B_n \cos 2n\pi t + B_n^i \sin 2n\pi t) \sin \frac{n\pi x}{2}$$

Initial velocity. $\frac{\partial u}{\partial t}(x, 0) = \sum 2n\pi B_n^i \sin \frac{n\pi x}{2} = 0$, so $B_n^i = 0$ for every n .

Initial displacement.

$$u(x, 0) = \sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{2} = 6 \sin \pi x - 3 \sin 4\pi x$$

$\sin \pi x = \sin \frac{2\pi x}{2}$ is the $n=2$ term and $\sin 4\pi x = \sin \frac{8\pi x}{2}$ is the $n=8$ term. Comparing coefficients (no Fourier integrals are needed):

$$B_2 = 6, B_8 = -3, B_n = 0 \text{ otherwise}$$

With $\lambda_2 = 4\pi$ and $\lambda_8 = 16\pi$:

| **Answer.** $u(x, t) = 6 \sin \pi x \cos 4\pi t - 3 \sin 4\pi x \cos 16\pi t$.

Check. $u(0, t) = u(2, t) = 0$ since $\sin 2\pi = \sin 8\pi = 0$; $u(x, 0) = 6 \sin \pi x - 3 \sin 4\pi x$; $u_t(x, 0) = 0$; and each term satisfies $u_{tt} = 16u_{xx}$, for example $(4\pi)^2 = 16\pi^2$. ✓

Question 6: Eigenvalues and Eigenvectors

(a) Eigenvalues and eigenvectors of a 3×3 matrix

Question 6(a). Find the eigenvalues and eigenvectors of the matrix (6 marks)

$$A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$$

Characteristic equation. Expanding $\det(A - \lambda I)$ along the first row,

$$(8 - \lambda)[(7 - \lambda)(3 - \lambda) - 16] + 6[-6(3 - \lambda) + 8] + 2[24 - 2(7 - \lambda)] = 0$$

$$(8 - \lambda)(\lambda^2 - 10\lambda + 5) + 6(6\lambda - 10) + 2(2\lambda + 10) = -\lambda^3 + 18\lambda^2 - 45\lambda = 0$$

$$\lambda(\lambda^2 - 18\lambda + 45) = \lambda(\lambda - 3)(\lambda - 15) = 0 \Rightarrow \lambda = 0, 3, 15$$

As a check, the eigenvalues add up to the trace: $0 + 3 + 15 = 8 + 7 + 3 = 18$. ✓

Eigenvectors.

$$\lambda = 0: \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix} x = 0. \text{ The cross product of the first two rows,}$$

$$(8, -6, 2) \times (-6, 7, -4) = (10, 20, 20), \text{ gives } x = (1, 2, 2)^T.$$

$$\lambda = 3: \begin{bmatrix} 5 & -6 & 2 \\ -6 & 4 & -4 \\ 2 & -4 & 0 \end{bmatrix} x = 0. \text{ The third row gives } x_1 = 2x_2; \text{ then the first gives}$$

$$10x_2 - 6x_2 + 2x_3 = 0, \text{ so } x_3 = -2x_2 \text{ and } x = (2, 1, -2)^T.$$

$$\lambda = 15: \begin{bmatrix} -7 & -6 & 2 \\ -6 & -8 & -4 \\ 2 & -4 & -12 \end{bmatrix} x = 0. \text{ The cross product of the first two rows,}$$

$$(-7, -6, 2) \times (-6, -8, -4) = (40, -40, 20), \text{ gives } x = (2, -2, 1)^T.$$

Check for $\lambda = 15$:

$A(2, -2, 1)^T = (16 + 12 + 2, -12 - 14 - 4, 4 + 8 + 3)^T = (30, -30, 15)^T = 15(2, -2, 1)^T$ ✓. The matrix is symmetric, so the eigenvectors are mutually orthogonal.

Answer. $\lambda_1 = 0$ with $x_1 = k(1, 2, 2)^T$; $\lambda_2 = 3$ with $x_2 = k(2, 1, -2)^T$; $\lambda_3 = 15$ with $x_3 = k(2, -2, 1)^T$ ($k \neq 0$).

(b) Principal directions of a stretched membrane

Question 6(b). An elastic membrane in the x_1x_2 -plane with boundary circle $x_1^2 + x_2^2 = 1$ is stretched so that a point $P:(x_1, x_2)$ goes over into the point $Q:(y_1, y_2)$ given by

$$y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = A x = \begin{bmatrix} 5 & 3 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Find the principal directions, that is, the directions of the position vector x of P for which the direction of the position vector y of Q is the same or exactly opposite. (6 marks)

We need the vectors x for which $y = Ax$ is parallel to x , that is $Ax = \lambda x$. These are the eigenvectors of

$$A = \begin{bmatrix} 5 & 3 \\ 3 & 5 \end{bmatrix}$$

Eigenvalues.

$$\det(A - \lambda I) = (5 - \lambda)^2 - 9 = \lambda^2 - 10\lambda + 16 = (\lambda - 8)(\lambda - 2) = 0 \Rightarrow \lambda_1 = 8, \lambda_2 = 2$$

Eigenvectors.

For $\lambda_1 = 8$: $-3x_1 + 3x_2 = 0 \Rightarrow x_2 = x_1$, so $x^{(1)} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, at 45° to the x_1 -axis.

For $\lambda_2 = 2$: $3x_1 + 3x_2 = 0 \Rightarrow x_2 = -x_1$, so $x^{(2)} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, at 135° (or -45°) to the x_1 -axis.

Both eigenvalues are positive, so in each principal direction y points the **same** way as x . The two directions are perpendicular because A is symmetric.

Interpretation. Along the 45° direction the membrane is stretched by a factor of 8, and along the 135° direction by a factor of 2. With coordinates u_1, u_2 along these principal directions, the boundary circle $x_1^2 + x_2^2 = 1$ becomes the ellipse

$$\frac{u_1^2}{8^2} + \frac{u_2^2}{2^2} = 1$$

Answer. The principal directions are $x_2 = x_1$ (at 45° , eigenvalue 8, stretch factor 8) and $x_2 = -x_1$ (at 135° , eigenvalue 2, stretch factor 2). The circle is deformed into an ellipse with semi-axes 8 and 2 along these directions.

Question 7: Differential Equations and Free Vibration

(a) Non-homogeneous second-order ODE

Question 7(a). Find the solution to the engineering problem represented by the differential equation below. (6 marks)

$$\frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} + 5y = x^2 + x e^{2x} + e^{-x} \sin 2x$$

Complementary function.

$$m^2 + 2m + 5 = 0 \Rightarrow m = -1 \pm 2i, y_c = e^{-x}(A \cos 2x + B \sin 2x)$$

Particular integral. $y_p = y_1 + y_2 + y_3$.

For x^2 . Try $y_1 = ax^2 + bx + c$:

$$2a + 2(2ax + b) + 5(ax^2 + bx + c) = 5ax^2 + (4a + 5b)x + (2a + 2b + 5c) = x^2$$

$$a = 1/5, b = -4a/5 = -4/25, c = -2a + 2b/5 = -2/125$$

For xe^{2x} . Put $y_2 = ue^{2x}$: $y_2'' + 2y_2' + 5y_2 = (u'' + 6u' + 13u)e^{2x} = xe^{2x}$. Try $u = px + q$:

$$6p + 13(px + q) = x \Rightarrow p = 1/13, q = -6/169:$$

$$y_2 = \frac{13x - 6}{169} e^{2x}$$

For $e^{-x} \sin 2x$. This term has the same form as y_c (since $-1 \pm 2i$ are the roots), so an extra factor of x is needed. Put $y_3 = ve^{-x}$; then $y_3' = (v' - v)e^{-x}$, $y_3'' = (v'' - 2v' + v)e^{-x}$ and

$$y_3'' + 2y_3' + 5y_3 = (v'' - 2v' + v + 2v' - 2v + 5v)e^{-x} = (v'' + 4v)e^{-x} = e^{-x} \sin 2x$$

Try $v = x(C \cos 2x + D \sin 2x)$. Then

$$v'' + 4v = 2 \frac{d}{dx}(C \cos 2x + D \sin 2x) = -4C \sin 2x + 4D \cos 2x = \sin 2x, \text{ so } C = -1/4, D = 0:$$

$$y_3 = -\frac{x}{4} e^{-x} \cos 2x$$

$$\textbf{Answer. } y = e^{-x}(A \cos 2x + B \sin 2x) + \frac{x^2}{5} - \frac{4x}{25} - \frac{2}{125} + \frac{13x - 6}{169} e^{2x} - \frac{x}{4} e^{-x} \cos 2x$$

(b) Undamped spring-mass system

Question 7(b). A spring with a mass of 2000 g has natural length of one-half a meter. A force of 25.6 N is required to maintain it stretched to a length of 700 mm. If the spring is stretched to a length of 700 mm and then released with initial velocity 0 m/s: (i) What will be the position of the mass at any time, t ? (ii) What length of travel (in m) would the mass achieve in $t = \pi/24$ seconds? (6 marks)

Spring constant. Hooke's law, $F = kx$, with the extension $x = 0.7 - 0.5 = 0.2$ m:

$$k(0.2) = 25.6 \Rightarrow k = 128 \text{ N/m}$$

The mass is $m = 2000 \text{ g} = 2 \text{ kg}$. Let $x(t)$ be the displacement of the mass from equilibrium (the natural length, 0.5 m), positive in the direction of stretching.

Equation of motion. There is no damping, so

$$m \frac{d^2 x}{dt^2} + kx = 0 \Rightarrow 2 \frac{d^2 x}{dt^2} + 128x = 0 \Rightarrow \frac{d^2 x}{dt^2} + 64x = 0$$

The auxiliary equation $r^2 + 64 = 0$ gives $r = \pm 8i$, so

$$x(t) = c_1 \cos 8t + c_2 \sin 8t$$

Initial conditions. The spring starts stretched to 0.7 m, so $x(0)=0.2$, and it is released from rest, so $x'(0)=0$.

$$x(0)=c_1=0.2, x'(0)=8c_2=0 \Rightarrow c_2=0$$

(i) Position at any time.

$$x(t)=0.2\cos 8t \text{ m}$$

The mass oscillates about the equilibrium position with amplitude 0.2 m, angular frequency $\omega=8$ rad/s and period $T=2\pi/8=\pi/4$ s.

(ii) At $t=\pi/24$ s.

$$x\left(\frac{\pi}{24}\right)=0.2\cos\left(\frac{8\pi}{24}\right)=0.2\cos\frac{\pi}{3}=0.2\times 0.5=0.1 \text{ m}$$

The velocity $x'(t)=-1.6\sin 8t$ stays negative for $0<t<\pi/8$, and $\pi/24<\pi/8$, so the mass moves steadily back towards equilibrium without turning. The distance travelled from the start is therefore

$$0.2-0.1=0.1 \text{ m}$$

and the spring's length at that moment is $0.5+0.1=0.6$ m.

Answer. (i) $x(t)=0.2\cos 8t$ m. (ii) At $t=\pi/24$ s the mass is 0.1 m from equilibrium, having travelled 0.1 m from its starting point (spring length 0.6 m).

Type 4

Question 2: Differential Equations and Free Vibration

(a) Non-homogeneous second-order ODE

Question 2(a). An engineering problem is represented by the differential equation below. Give the solution to the problem. (6 marks)

$$\frac{d^2 y}{dx^2} + 4 \frac{dy}{dx} + 4y = x^2 + x e^{2x} + e^{-2x} \sin 2x$$

Complementary function.

$$m^2 + 4m + 4 = (m+2)^2 = 0 \Rightarrow m = -2 \text{ (repeated)}, y_c = (A+Bx)e^{-2x}$$

Particular integral. $y_p = y_1 + y_2 + y_3$.

For x^2 . Try $y_1 = ax^2 + bx + c$:

$$2a + 4(2ax + b) + 4(ax^2 + bx + c) = 4ax^2 + (8a + 4b)x + (2a + 4b + 4c) = x^2$$

$$a = 1/4, b = -2a = -1/2, c = -2a + 4b/4 = 3/8$$

For $x e^{2x}$. Put $y_2 = u e^{2x}$: $y_2'' + 4y_2' + 4y_2 = (u'' + 8u' + 16u)e^{2x} = x e^{2x}$. Try $u = px + q$:

$$8p + 16(px + q) = x \Rightarrow p = 1/16, q = -1/32:$$

$$y_2 = \frac{2x-1}{32} e^{2x}$$

For $e^{-2x} \sin 2x$. Put $y_3 = v e^{-2x}$. Because -2 is a double root of the auxiliary equation, the operator reduces to v'' :

$$y_3'' + 4y_3' + 4y_3 = (v'' - 4v' + 4v + 4v' - 8v + 4v)e^{-2x} = v'' e^{-2x} = e^{-2x} \sin 2x$$

So $v'' = \sin 2x$, $v = -1/4 \sin 2x$ and $y_3 = -1/4 e^{-2x} \sin 2x$.

$$\text{Answer. } y = (A+Bx)e^{-2x} + \frac{x^2}{4} - \frac{x}{2} + \frac{3}{8} + \frac{2x-1}{32} e^{2x} - \frac{1}{4} e^{-2x} \sin 2x$$

(b) Undamped spring-mass system

Question 2(b). A spring with a mass of 2000 g has natural length of one-half a meter. A force of 25.6 N is required to maintain it stretched to a length of 700 mm. If the spring is stretched to a length of 700 mm and then released with initial velocity 0 m/s: (i) What will be the position of the mass at any time, t ? (ii) What length of travel (in m) would the mass achieve in $t = \pi/24$ seconds? (6 marks)

Spring constant. Hooke's law, $F = kx$, with the extension $x = 0.7 - 0.5 = 0.2$ m:

$$k(0.2)=25.6 \Rightarrow k=128 \text{ N/m}$$

The mass is $m=2000 \text{ g}=2 \text{ kg}$. Let $x(t)$ be the displacement of the mass from equilibrium (the natural length, 0.5 m), positive in the direction of stretching.

Equation of motion. There is no damping, so

$$m \frac{d^2 x}{dt^2} + kx = 0 \Rightarrow 2 \frac{d^2 x}{dt^2} + 128x = 0 \Rightarrow \frac{d^2 x}{dt^2} + 64x = 0$$

The auxiliary equation $r^2 + 64 = 0$ gives $r = \pm 8i$, so

$$x(t) = c_1 \cos 8t + c_2 \sin 8t$$

Initial conditions. The spring starts stretched to 0.7 m , so $x(0) = 0.2$, and it is released from rest, so $x'(0) = 0$.

$$x(0) = c_1 = 0.2, x'(0) = 8c_2 = 0 \Rightarrow c_2 = 0$$

(i) Position at any time.

$$x(t) = 0.2 \cos 8t \text{ m}$$

The mass oscillates about the equilibrium position with amplitude 0.2 m , angular frequency $\omega = 8 \text{ rad/s}$ and period $T = 2\pi/8 = \pi/4 \text{ s}$.

(ii) At $t = \pi/24 \text{ s}$.

$$x\left(\frac{\pi}{24}\right) = 0.2 \cos\left(\frac{8\pi}{24}\right) = 0.2 \cos \frac{\pi}{3} = 0.2 \times 0.5 = 0.1 \text{ m}$$

The velocity $x'(t) = -1.6 \sin 8t$ stays negative for $0 < t < \pi/8$, and $\pi/24 < \pi/8$, so the mass moves steadily back towards equilibrium without turning. The distance travelled from the start is therefore

$$0.2 - 0.1 = 0.1 \text{ m}$$

and the spring's length at that moment is $0.5 + 0.1 = 0.6 \text{ m}$.

Answer. (i) $x(t) = 0.2 \cos 8t \text{ m}$. (ii) At $t = \pi/24 \text{ s}$ the mass is 0.1 m from equilibrium, having travelled 0.1 m from its starting point (spring length 0.6 m).

Question 3: Numerical Differentiation

(a) Rate of population growth

Question 3(a). The table below gives the census population (in millions) of a state for the years 1961 to 2001. Find the rate of growth of the population in the year 2001. (6 marks)

Year (x)	1961	1971	1981	1991	2001
Population (y)	19.96	36.65	58.81	77.21	94.61

The year 2001 is at the **end** of the table, so we use Newton's backward difference formula with $h=10$ years.

Backward difference table.

Year x	y	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$
1961	19.96				
1971	36.65	16.69			
1981	58.81	22.16	5.47		
1991	77.21	18.40	-3.76	-9.23	
2001	94.61	17.40	-1.00	2.76	11.99

Derivative at the last point $x_n=2001$ (where $p=0$):

$$\left(\frac{dy}{dx}\right)_{x_n} = \frac{1}{h} \left[\nabla y_n + \frac{1}{2} \nabla^2 y_n + \frac{1}{3} \nabla^3 y_n + \frac{1}{4} \nabla^4 y_n \right]$$

$$\therefore \frac{1}{10} \left[17.40 + \frac{-1.00}{2} + \frac{2.76}{3} + \frac{11.99}{4} \right] = \frac{1}{10} [17.40 - 0.50 + 0.92 + 2.9975] = \frac{20.8175}{10} = 2.0818$$

| **Answer.** In 2001 the population was growing at about **2.08 million per year**.

(b) Derivatives from tabulated data

Question 3(b). Given the following data points, determine $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ at $x=1.1$ and $x=1.5$. (6 marks)

x	1.0	1.1	1.2	1.3	1.4	1.5	1.6
$f(x)$	7.989	8.403	8.781	9.129	9.451	9.750	10.031

Here $h=0.1$. The point $x=1.1$ is near the start of the table, so we use Newton's **forward** formula there; $x=1.5$ is near the end, so we use the **backward** formula.

Difference table (row x_i lists $\Delta^k f_i$, the forward differences starting at that row).

x	$f(x)$	Δ	Δ^2	Δ^3	Δ^4	Δ^5	Δ^6
1.0	7.989	0.414	-0.036	0.006	-0.002	0.001	0.002
1.1	8.403	0.378	-0.030	0.004	-0.001	0.003	
1.2	8.781	0.348	-0.026	0.003	0.002		
1.3	9.129	0.322	-0.023	0.005			
1.4	9.451	0.299	-0.018				
1.5	9.750	0.281					

x	$f(x)$	Δ	Δ^2	Δ^3	Δ^4	Δ^5	Δ^6
1.6	10.031						

At $x=1.1$ (forward differences, $x_0=1.1$)

From the table, $\Delta y_0=0.378$, $\Delta^2 y_0=-0.030$, $\Delta^3 y_0=0.004$, $\Delta^4 y_0=-0.001$ and $\Delta^5 y_0=0.003$

$$\left(\frac{dy}{dx}\right)_{x_0} = \frac{1}{h} \left[\Delta y_0 - \frac{1}{2} \Delta^2 y_0 + \frac{1}{3} \Delta^3 y_0 - \frac{1}{4} \Delta^4 y_0 + \frac{1}{5} \Delta^5 y_0 \right]$$

$$\therefore \frac{1}{0.1} \left[0.378 + \frac{0.030}{2} + \frac{0.004}{3} + \frac{0.001}{4} + \frac{0.003}{5} \right] = 10 [0.378 + 0.015 + 0.00133 + 0.00025 + 0.0006] = 3.952$$

$$\left(\frac{d^2y}{dx^2}\right)_{x_0} = \frac{1}{h^2} \left[\Delta^2 y_0 - \Delta^3 y_0 + \frac{11}{12} \Delta^4 y_0 - \frac{5}{6} \Delta^5 y_0 \right]$$

$$\therefore 100 \left[-0.030 - 0.004 + \frac{11}{12}(-0.001) - \frac{5}{6}(0.003) \right] = 100 [-0.030 - 0.004 - 0.00092 - 0.0025] = -3.742$$

At $x=1.5$ (backward differences, $x_n=1.5$)

From the table, $\nabla y_n=0.299$, $\nabla^2 y_n=-0.023$, $\nabla^3 y_n=0.003$, $\nabla^4 y_n=-0.001$ and $\nabla^5 y_n=0.001$.

$$\left(\frac{dy}{dx}\right)_{x_n} = \frac{1}{h} \left[\nabla y_n + \frac{1}{2} \nabla^2 y_n + \frac{1}{3} \nabla^3 y_n + \frac{1}{4} \nabla^4 y_n + \frac{1}{5} \nabla^5 y_n \right]$$

$$\therefore 10 [0.299 - 0.0115 + 0.001 - 0.00025 + 0.0002] = 2.885$$

$$\left(\frac{d^2y}{dx^2}\right)_{x_n} = \frac{1}{h^2} \left[\nabla^2 y_n + \nabla^3 y_n + \frac{11}{12} \nabla^4 y_n + \frac{5}{6} \nabla^5 y_n \right]$$

$$\therefore 100 [-0.023 + 0.003 - 0.00092 + 0.00083] = -2.008$$

Answer. At $x=1.1$: $\frac{dy}{dx} \approx 3.952$ and $\frac{d^2y}{dx^2} \approx -3.742$. At $x=1.5$: $\frac{dy}{dx} \approx 2.885$ and $\frac{d^2y}{dx^2} \approx -2.008$.

Question 4: Series Solutions and Bessel's Equation

(a) Power series solution

Question 4(a). Find the power series solution of $4(y'' + y) = 0$. (6 marks)

Dividing by the non-zero constant 4 leaves $y'' + y = 0$.

Assume a power series solution about $x=0$:

$$y = \sum_{n=0}^{\infty} a_n x^n, y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} = \sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n$$

Substituting into $y'' + y = 0$:

$$\sum_{n=0}^{\infty} [(n+2)(n+1) a_{n+2} + a_n] x^n = 0$$

Every coefficient must vanish, which gives the **recurrence relation**

$$a_{n+2} = -\frac{a_n}{(n+2)(n+1)}, n=0, 1, 2, \dots$$

Even coefficients (in terms of a_0):

$$a_2 = -\frac{a_0}{2!}, a_4 = -\frac{a_2}{4 \cdot 3} = \frac{a_0}{4!}, a_6 = -\frac{a_4}{6 \cdot 5} = -\frac{a_0}{6!}, \dots$$

Odd coefficients (in terms of a_1):

$$a_3 = -\frac{a_1}{3!}, a_5 = -\frac{a_3}{5 \cdot 4} = \frac{a_1}{5!}, a_7 = -\frac{a_1}{7!}, \dots$$

Hence

$$y = a_0 \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots \right) + a_1 \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots \right)$$

The two series are the Maclaurin series of $\cos x$ and $\sin x$, and both converge for all x .

$$\left| \textbf{Answer. } y = a_0 \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots \right) + a_1 \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots \right) = a_0 \cos x + a_1 \sin x, \text{ where } a_0 \text{ and } a_1 \text{ are arbitrary constants.} \right|$$

(b) Bessel's equation

Question 4(b). Find the general solution, in terms of Bessel functions, of the equation $x y'' + y' + \left(\frac{4}{x}\right) y = 0$. (6 marks)

The fraction in the equation reduces to $1/2$, so the equation is

$$x y'' + y' + 1/2 y = 0$$

Change of variable. Equations of the form $x y'' + y' + \lambda y = 0$ become Bessel's equation under $z = 2\sqrt{\lambda x}$. Here $\lambda = 1/2$, so put

$$z = 2\sqrt{x/2} = \sqrt{2x}, z^2 = 2x, \frac{dz}{dx} = \frac{1}{z}$$

Then

$$\frac{dy}{dx} = \frac{1}{z} \frac{dy}{dz}, \frac{d^2 y}{dx^2} = \frac{1}{z} \frac{d}{dz} \left(\frac{1}{z} \frac{dy}{dz} \right) = \frac{1}{z^2} \frac{d^2 y}{dz^2} - \frac{1}{z^3} \frac{dy}{dz}$$

With $x = z^2/2$:

$$x \frac{d^2 y}{dx^2} + \frac{dy}{dx} + \frac{1}{2}y = \frac{1}{2} \frac{d^2 y}{dz^2} - \frac{1}{2z} \frac{dy}{dz} + \frac{1}{z} \frac{dy}{dz} + \frac{1}{2}y = \frac{1}{2} \left(\frac{d^2 y}{dz^2} + \frac{1}{z} \frac{dy}{dz} + y \right) = 0$$

Multiplying by $2z^2$:

$$z^2 \frac{d^2 y}{dz^2} + z \frac{dy}{dz} + (z^2 - \nu^2)y = 0 \text{ with } \nu = 0$$

This is **Bessel's equation of order** $\nu = 0$.

General solution. For order $\nu = 0$ the two independent solutions are $J_0(z)$ and the Bessel function of the second kind, $Y_0(z)$:

$$y = c_1 J_0(z) + c_2 Y_0(z) = c_1 J_0(\sqrt{2x}) + c_2 Y_0(\sqrt{2x})$$

In series form, the first solution is

$$J_0(\sqrt{2x}) = \sum_{m=0}^{\infty} \frac{(-1)^m}{(m!)^2} \left(\frac{x}{2} \right)^m = 1 - \frac{x}{2} + \frac{x^2}{16} - \frac{x^3}{288} + \dots$$

Answer. $y = c_1 J_0(\sqrt{2x}) + c_2 Y_0(\sqrt{2x})$.

Question 5: Eigenvalues and Eigenvectors

(a) Eigenvalues and eigenvectors of a 3×3 matrix

Question 5(a). Find the eigenvalues and eigenvectors of the matrix (6 marks)

$$A = \begin{bmatrix} 1 & 1 & -2 \\ -1 & 0 & 1 \\ -2 & 1 & 1 \end{bmatrix}$$

Characteristic equation. Expanding $\det(A - \lambda I)$ along the first row,

$$(1 - \lambda)[- \lambda(1 - \lambda) - 1] - 1[-(1 - \lambda) + 2] - 2[-1 - 2\lambda]$$

$$= (1 - \lambda)(\lambda^2 - \lambda - 1) - (1 + \lambda) + 2 + 4\lambda = -\lambda^3 + 2\lambda^2 + 3\lambda = 0$$

$$\lambda(\lambda^2 - 2\lambda - 3) = \lambda(\lambda - 3)(\lambda + 1) = 0 \Rightarrow \lambda = -1, 0, 3$$

As a check, $-1 + 0 + 3 = 2 = 1 + 0 + 1$, the trace. ✓

Eigenvectors.

$$\lambda = -1: \begin{bmatrix} 2 & 1 & -2 \\ -1 & 1 & 1 \\ -2 & 1 & 2 \end{bmatrix} x = 0. \text{ Adding rows 1 and 3 gives } 2x_2 = 0, \text{ so } x_2 = 0 \text{ and } x_1 = x_3:$$

$$x = (1, 0, 1)^T.$$

$$\lambda=0: \begin{bmatrix} 1 & 1 & -2 \\ -1 & 0 & 1 \\ -2 & 1 & 1 \end{bmatrix} x=0. \text{ Row 2 gives } x_1=x_3; \text{ row 1 then gives } x_2=x_3: x=(1,1,1)^T.$$

$$\lambda=3: \begin{bmatrix} -2 & 1 & -2 \\ -1 & -3 & 1 \\ -2 & 1 & -2 \end{bmatrix} x=0. \text{ Row 1 gives } x_2=2x_1+2x_3; \text{ row 2 then gives}$$

$$-x_1-6x_1-6x_3+x_3=0, \text{ so } 7x_1=-5x_3. \text{ Taking } x_3=-7: x=(5,-4,-7)^T.$$

Check for $\lambda=3$:

$$A(5,-4,-7)^T = (5-4+14, -5-7, -10-4-7)^T = (15, -12, -21)^T = 3(5,-4,-7)^T \checkmark$$

Answer. $\lambda_1=-1$ with $x_1=k(1,0,1)^T$; $\lambda_2=0$ with $x_2=k(1,1,1)^T$; $\lambda_3=3$ with $x_3=k(5,-4,-7)^T$ ($k \neq 0$).

(b) Principal directions of a stretched membrane

Question 5(b). An elastic membrane in the x_1x_2 -plane with boundary circle $x_1^2+x_2^2=1$ is stretched so that a point $P:(x_1,x_2)$ goes over into the point $Q:(y_1,y_2)$ given by

$$y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = A x = \begin{bmatrix} 5 & 3 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Find the principal directions, that is, the directions of the position vector x of P for which the direction of the position vector y of Q is the same or exactly opposite. (6 marks)

We need the vectors x for which $y=Ax$ is parallel to x , that is $Ax=\lambda x$. These are the eigenvectors of

$$A = \begin{bmatrix} 5 & 3 \\ 3 & 5 \end{bmatrix}$$

Eigenvalues.

$$\det(A - \lambda I) = (5 - \lambda)^2 - 9 = \lambda^2 - 10\lambda + 16 = (\lambda - 8)(\lambda - 2) = 0 \Rightarrow \lambda_1=8, \lambda_2=2$$

Eigenvectors.

For $\lambda_1=8$: $-3x_1+3x_2=0 \Rightarrow x_2=x_1$, so $x^{(1)} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, at 45° to the x_1 -axis.

For $\lambda_2=2$: $3x_1+3x_2=0 \Rightarrow x_2=-x_1$, so $x^{(2)} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, at 135° (or -45°) to the x_1 -axis.

Both eigenvalues are positive, so in each principal direction y points the **same** way as x . The two directions are perpendicular because A is symmetric.

Interpretation. Along the 45° direction the membrane is stretched by a factor of 8, and along the 135° direction by a factor of 2. With coordinates u_1, u_2 along these principal directions, the boundary circle $x_1^2+x_2^2=1$ becomes the ellipse

$$\frac{u_1^2}{8^2} + \frac{u_2^2}{2^2} = 1$$

Answer. The principal directions are $x_2 = x_1$ (at 45° , eigenvalue 8, stretch factor 8) and $x_2 = -x_1$ (at 135° , eigenvalue 2, stretch factor 2). The circle is deformed into an ellipse with semi-axes 8 and 2 along these directions.

Question 6: Legendre's Equation and the Wave Equation

(a) Legendre's equation

Question 6(a). Solve this special Legendre equation, $(1-x^2)y'' - 2xy' + 2y = 0$, which occurs in models exhibiting spherical symmetry. (6 marks)

This is Legendre's equation $(1-x^2)y'' - 2xy' + n(n+1)y = 0$ with $n(n+1) = 2$, that is $n = 1$.

Power series. Put $y = \sum_{m=0}^{\infty} a_m x^m$. Then

$$(1-x^2) \sum_m m(m-1)a_m x^{m-2} - 2x \sum_m m a_m x^{m-1} + 2 \sum_m a_m x^m = 0$$

Collecting the coefficient of x^m :

$$(m+2)(m+1)a_{m+2} - [m(m-1) + 2m - 2]a_m = 0$$

Since $m(m-1) + 2m - 2 = m^2 + m - 2 = (m+2)(m-1)$, the recurrence relation is

$$a_{m+2} = \frac{(m+2)(m-1)}{(m+2)(m+1)} a_m = \frac{m-1}{m+1} a_m$$

Odd series ($m=1$): $a_3 = 0 \cdot a_1 = 0$, so $a_5 = a_7 = \dots = 0$ and the series stops:

$$y_1 = a_1 x = a_1 P_1(x)$$

Even series (from a_0):

$$a_2 = -a_0, a_4 = \frac{1}{3}a_2 = -\frac{a_0}{3}, a_6 = \frac{3}{5}a_4 = -\frac{a_0}{5}, a_8 = -\frac{a_0}{7}, \dots$$

$$y_2 = a_0 \left(1 - x^2 - \frac{x^4}{3} - \frac{x^6}{5} - \frac{x^8}{7} - \dots \right) = a_0 \left(1 - \frac{x}{2} \ln \frac{1+x}{1-x} \right), \forall x \in (-1, 1)$$

because $\frac{1}{2} \ln \frac{1+x}{1-x} = x + \frac{x^3}{3} + \frac{x^5}{5} + \dots$.

Check of $y_1 = x$: $(1-x^2)(0) - 2x(1) + 2x = 0$. ✓

Answer. $y = c_1 x + c_2 \left(1 - x^2 - \frac{x^4}{3} - \frac{x^6}{5} - \dots \right) = c_1 P_1(x) + c_2 \left(1 - \frac{x}{2} \ln \frac{1+x}{1-x} \right)$ for $|x| < 1$. The polynomial solution is the Legendre polynomial $P_1(x) = x$.

(b) One-dimensional wave equation

Question 6(b). Find the solution to this one-dimensional wave equation:

$\frac{\partial^2 u}{\partial x^2} = \frac{1}{16} \frac{\partial^2 u}{\partial t^2}$ for $0 < x < 2, t > 0$. The boundary conditions are $u(0, t) = u(2, t) = 0$. The initial conditions are (i) $u(x, 0) = 6 \sin \pi x - 3 \sin 4 \pi x$ and (ii) $\frac{\partial u}{\partial t}(x, 0) = 0$. (6 marks)

The equation is

$$\frac{\partial^2 u}{\partial x^2} = \frac{1}{16} \frac{\partial^2 u}{\partial t^2} \Leftrightarrow \frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}, c^2 = 16, c = 4, L = 2$$

Separation of variables. Put $u = F(x)G(t)$:

$$\frac{F''}{F} = \frac{G''}{16G} = -p^2$$

so $F'' + p^2 F = 0$ and $G'' + 16p^2 G = 0$.

Boundary conditions. $u(0, t) = u(2, t) = 0$ need $F(0) = F(2) = 0$. With $F = A \cos px + B \sin px$, $F(0) = 0$ gives $A = 0$, and $\sin 2p = 0$ gives

$$p = \frac{n\pi}{2}, F_n(x) = \sin \frac{n\pi x}{2}, n = 1, 2, 3, \dots$$

Time part. $G_n'' + \lambda_n^2 G_n = 0$ with $\lambda_n = \frac{cn\pi}{L} = \frac{4n\pi}{2} = 2n\pi$:

$$G_n(t) = B_n \cos 2n\pi t + B_n^i \sin 2n\pi t$$

so

$$u(x, t) = \sum_{n=1}^{\infty} (B_n \cos 2n\pi t + B_n^i \sin 2n\pi t) \sin \frac{n\pi x}{2}$$

Initial velocity. $\frac{\partial u}{\partial t}(x, 0) = \sum 2n\pi B_n^i \sin \frac{n\pi x}{2} = 0$, so $B_n^i = 0$ for every n .

Initial displacement.

$$u(x, 0) = \sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{2} = 6 \sin \pi x - 3 \sin 4\pi x$$

$\sin \pi x = \sin \frac{2\pi x}{2}$ is the $n=2$ term and $\sin 4\pi x = \sin \frac{8\pi x}{2}$ is the $n=8$ term. Comparing coefficients (no Fourier integrals are needed):

$$B_2 = 6, B_8 = -3, B_n = 0 \text{ otherwise}$$

With $\lambda_2 = 4\pi$ and $\lambda_8 = 16\pi$:

| **Answer.** $u(x, t) = 6 \sin \pi x \cos 4\pi t - 3 \sin 4\pi x \cos 16\pi t$.

Check. $u(0, t) = u(2, t) = 0$ since $\sin 2\pi = \sin 8\pi = 0$; $u(x, 0) = 6 \sin \pi x - 3 \sin 4\pi x$; $u_t(x, 0) = 0$; and each term satisfies $u_{tt} = 16u_{xx}$, for example $(4\pi)^2 = 16\pi^2$. ✓

Question 7: RLC Circuit

Question 7. Consider an RLC circuit with a $17/50\Omega$ resistor, a $1/100$ H inductor and a $100/93$ F capacitor, driven by the voltage $E(t)=0.06\sin 3t$ V. (a) Write the differential equation associated with this circuit in terms of the current I . (6 marks) (b) If the initial charge and initial current on the capacitor are both zero, find the current I and the voltage across the resistor E_R in terms of time t . (6 marks)

(a) Differential equation in terms of I

Kirchhoff's voltage law, with $q=\int I dt$, gives

$$L \frac{dI}{dt} + RI + \frac{1}{C} \int I dt = E(t)$$

Differentiating with respect to t removes the integral:

$$L \frac{d^2 I}{dt^2} + R \frac{dI}{dt} + \frac{1}{C} I = \frac{dE}{dt}$$

With $L=1/100$, $R=17/50$, $1/C=93/100$ and $\frac{dE}{dt}=0.18\cos 3t$:

$$\frac{1}{100} \frac{d^2 I}{dt^2} + \frac{17}{50} \frac{dI}{dt} + \frac{93}{100} I = 0.18\cos 3t$$

Multiplying by 100:

$$\frac{d^2 I}{dt^2} + 34 \frac{dI}{dt} + 93 I = 18\cos 3t$$

(b) Current and resistor voltage

Initial conditions. $I(0)=0$. At $t=0$ the circuit equation gives

$LI'(0)=E(0)-RI(0)-q(0)/C=0-0-0$, so $I'(0)=0$.

Complementary function. $r^2+34r+93=0$ gives $r=\frac{-34\pm\sqrt{1156-372}}{2}=\frac{-34\pm 28}{2}$, so

$r=-3$ or $r=-31$:

$$I_c=c_1 e^{-3t}+c_2 e^{-31t}$$

Particular integral. Try $I_p=A\cos 3t+B\sin 3t$, with $I_p''=-9I_p$:

$$(93-9)(A\cos 3t+B\sin 3t)+34(-3A\sin 3t+3B\cos 3t)=18\cos 3t$$

$$\cos 3t: 84A+102B=18, \sin 3t: 84B-102A=0$$

The second gives $B=17/14A$. Substituting, $84A+1734/14A=18$, so $2910/14A=18$, giving $A=42/485$ and $B=51/485$.

Constants.

$$I(0) = c_1 + c_2 + \frac{42}{485} = 0, I'(0) = -3c_1 - 31c_2 + \frac{153}{485} = 0$$

Solving: $c_1 = -3/28$ and $c_2 = 279/13580$.

Current.

$$I(t) = -\frac{3}{28}e^{-3t} + \frac{279}{13580}e^{-31t} + \frac{42 \cos 3t + 51 \sin 3t}{485}$$

$$I(t) \approx -0.1071e^{-3t} + 0.0205e^{-31t} + 0.0866 \cos 3t + 0.1052 \sin 3t \text{ A}$$

Voltage across the resistor. $E_R = RI = 17/50 I$:

$$E_R(t) = -\frac{51}{1400}e^{-3t} + \frac{4743}{679000}e^{-31t} + \frac{357}{12125} \cos 3t + \frac{867}{24250} \sin 3t$$

$$E_R(t) \approx -0.0364e^{-3t} + 0.0070e^{-31t} + 0.0294 \cos 3t + 0.0358 \sin 3t \text{ V}$$

Answer. (a) $I'' + 34I' + 93I = 18 \cos 3t$. (b)

$I = -3/28 e^{-3t} + 279/13580 e^{-31t} + 1/485 (42 \cos 3t + 51 \sin 3t)$ A, and

$E_R = 17/50 I \approx -0.0364 e^{-3t} + 0.0070 e^{-31t} + 0.0294 \cos 3t + 0.0358 \sin 3t$ V.

Type 5

Question 2: Differential Equations and Free Vibration

(a) Non-homogeneous second-order ODE

Question 2(a). Find the solution to the engineering problem represented by the differential equation below. (6 marks)

$$\frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} + 5y = x^2 + x e^{2x} + e^{-x} \cos 2x$$

Complementary function.

$$m^2 + 2m + 5 = 0 \Rightarrow m = -1 \pm 2i, y_c = e^{-x} (A \cos 2x + B \sin 2x)$$

Particular integral. $y_p = y_1 + y_2 + y_3$.

For x^2 . Try $y_1 = ax^2 + bx + c$:

$$5ax^2 + (4a + 5b)x + (2a + 2b + 5c) = x^2 \Rightarrow a = 1/5, b = -4/25, c = -2/125$$

For $x e^{2x}$. Put $y_2 = u e^{2x}$: $(u'' + 6u' + 13u)e^{2x} = x e^{2x}$. Try $u = px + q$:

$$6p + 13(px + q) = x \Rightarrow p = 1/13, q = -6/169:$$

$$y_2 = \frac{13x - 6}{169} e^{2x}$$

For $e^{-x} \cos 2x$. This term has the same form as y_c , so an extra factor of x is needed.

Put $y_3 = v e^{-x}$; then

$$y_3'' + 2y_3' + 5y_3 = (v'' - 2v' + v + 2v' - 2v + 5v)e^{-x} = (v'' + 4v)e^{-x} = e^{-x} \cos 2x$$

Try $v = x(C \cos 2x + D \sin 2x)$. Then

$$v'' + 4v = 2 \frac{d}{dx} (C \cos 2x + D \sin 2x) = -4C \sin 2x + 4D \cos 2x = \cos 2x, \text{ so } C = 0, D = 1/4:$$

$$y_3 = \frac{x}{4} e^{-x} \sin 2x$$

$$\text{Answer. } y = e^{-x} (A \cos 2x + B \sin 2x) + \frac{x^2}{5} - \frac{4x}{25} - \frac{2}{125} + \frac{13x - 6}{169} e^{2x} + \frac{x}{4} e^{-x} \sin 2x$$

(b) Undamped spring-mass system

Question 2(b). A spring with a mass of 2000 g has natural length of one-half a meter. A force of 25.6 N is required to maintain it stretched to a length of 700 mm. If the spring is stretched to a length of 700 mm and then released with initial velocity 0 m/s: (i) What will be the position of the mass at any time, t ? (ii) What length of travel (in m) would the mass achieve in $t = \pi/24$ seconds? (6 marks)

Spring constant. Hooke's law, $F=kx$, with the extension $x=0.7-0.5=0.2$ m:

$$k(0.2)=25.6 \Rightarrow k=128 \text{ N/m}$$

The mass is $m=2000 \text{ g}=2 \text{ kg}$. Let $x(t)$ be the displacement of the mass from equilibrium (the natural length, 0.5 m), positive in the direction of stretching.

Equation of motion. There is no damping, so

$$m \frac{d^2 x}{dt^2} + kx = 0 \Rightarrow 2 \frac{d^2 x}{dt^2} + 128x = 0 \Rightarrow \frac{d^2 x}{dt^2} + 64x = 0$$

The auxiliary equation $r^2+64=0$ gives $r=\pm 8i$, so

$$x(t)=c_1 \cos 8t + c_2 \sin 8t$$

Initial conditions. The spring starts stretched to 0.7 m, so $x(0)=0.2$, and it is released from rest, so $x'(0)=0$.

$$x(0)=c_1=0.2, x'(0)=8c_2=0 \Rightarrow c_2=0$$

(i) Position at any time.

$$x(t)=0.2 \cos 8t \text{ m}$$

The mass oscillates about the equilibrium position with amplitude 0.2 m, angular frequency $\omega=8 \text{ rad/s}$ and period $T=2\pi/8=\pi/4 \text{ s}$.

(ii) At $t=\pi/24 \text{ s}$.

$$x\left(\frac{\pi}{24}\right)=0.2 \cos\left(\frac{8\pi}{24}\right)=0.2 \cos \frac{\pi}{3}=0.2 \times 0.5=0.1 \text{ m}$$

The velocity $x'(t)=-1.6 \sin 8t$ stays negative for $0 < t < \pi/8$, and $\pi/24 < \pi/8$, so the mass moves steadily back towards equilibrium without turning. The distance travelled from the start is therefore

$$0.2 - 0.1 = 0.1 \text{ m}$$

and the spring's length at that moment is $0.5+0.1=0.6 \text{ m}$.

Answer. (i) $x(t)=0.2 \cos 8t \text{ m}$. (ii) At $t=\pi/24 \text{ s}$ the mass is 0.1 m from equilibrium, having travelled 0.1 m from its starting point (spring length 0.6 m).

Question 3: RLC Circuit

Question 3. A series RLC circuit consists of a 5Ω resistor, a 0.1 H inductor and a $1000 \mu\text{F}$ capacitor. The circuit starts with zero charge on the capacitor and zero current through the inductor. (a) Find the charge on the capacitor as a function of time (i) when the circuit is connected to a 6 V DC source, and (ii) when the circuit is driven by an AC source given by $E(t)=6 \sin 2t$. (6 marks) (b) For each case in part (a), determine the corresponding current in the circuit. (6 marks)

The circuit equation. With $C=1000 \mu\text{F}=10^{-3} \text{ F}$, so $1/C=1000$:

$$0.1 \frac{d^2 q}{dt^2} + 5 \frac{dq}{dt} + 1000 q = E(t) \Rightarrow \frac{d^2 q}{dt^2} + 50 \frac{dq}{dt} + 10000 q = 10 E(t)$$

with $q(0)=0$ and $q'(0)=I(0)=0$.

Complementary function. $r^2 + 50r + 10000 = 0$ gives

$$r = -25 \pm \sqrt{625 - 10000} = -25 \pm i\sqrt{9375} = -25 \pm 25\sqrt{15}i$$

Write $\omega = 25\sqrt{15} \approx 96.82$ rad/s. Then

$$q_c = e^{-25t} (c_1 \cos \omega t + c_2 \sin \omega t)$$

Case (i): 6 V DC source

Particular integral. $10000 q_p = 60$, so $q_p = 0.006$.

Constants. $q(0) = c_1 + 0.006 = 0 \Rightarrow c_1 = -0.006$.

$$q'(0) = -25c_1 + \omega c_2 = 0 \Rightarrow c_2 = \frac{25c_1}{\omega} = -\frac{0.006}{\sqrt{15}} = -\frac{\sqrt{15}}{2500} \approx -0.001549.$$

$$q(t) = 0.006 \left[1 - e^{-25t} \left(\cos \omega t + \frac{1}{\sqrt{15}} \sin \omega t \right) \right]$$

Current. Differentiating (the cosine terms cancel),

$$I(t) = 0.006 \left(\omega + \frac{625}{\omega} \right) e^{-25t} \sin \omega t = \frac{60}{\omega} e^{-25t} \sin \omega t = \frac{4\sqrt{15}}{25} e^{-25t} \sin \omega t \approx 0.6197 e^{-25t} \sin(96.82t)$$

Case (ii): AC source, $E = 6 \sin 2t$

The right-hand side is $10E = 60 \sin 2t$.

Particular integral. Try $q_p = A \cos 2t + B \sin 2t$, with $q_p'' = -4q_p$:

$$\cos 2t: 9996A + 100B = 0, \sin 2t: 9996B - 100A = 60$$

Solving,

$$B = \frac{60 \times 9996}{9996^2 + 100^2} = \frac{37485}{6245626} \approx 6.0018 \times 10^{-3}, A = -\frac{100B}{9996} = -\frac{375}{6245626} \approx -6.004 \times 10^{-5}$$

Constants. $q(0) = c_1 + A = 0 \Rightarrow c_1 = 6.004 \times 10^{-5}$.

$$q'(0) = -25c_1 + \omega c_2 + 2B = 0 \Rightarrow c_2 = \frac{25c_1 - 2B}{\omega} \approx -1.0847 \times 10^{-4}.$$

$$q(t) \approx e^{-25t} (6.004 \times 10^{-5} \cos \omega t - 1.0847 \times 10^{-4} \sin \omega t) + 6.0018 \times 10^{-3} \sin 2t - 6.004 \times 10^{-5} \cos 2t$$

Current.

$$I(t) \approx -e^{-25t} (0.012004 \cos \omega t + 0.003102 \sin \omega t) + 0.012004 \cos 2t + 0.000120 \sin 2t$$

Check: $I(0) = -0.012004 + 0.012004 = 0$. ✓ The transient dies away within about 0.2 s (since e^{-25t} is then below 1 %), leaving the steady-state current $\approx 0.0120 \cos 2t$ A.

Answer. (i) $q = 0.006 \left[1 - e^{-25t} (\cos 25\sqrt{15}t + 1/\sqrt{15} \sin 25\sqrt{15}t) \right]$ C and

$I = 4\sqrt{15}/25 e^{-25t} \sin 25\sqrt{15}t \approx 0.620 e^{-25t} \sin 96.82t$ A. (ii)

$q \approx e^{-25t} (6.00 \times 10^{-5} \cos 96.82t - 1.08 \times 10^{-4} \sin 96.82t) + 6.00 \times 10^{-3} \sin 2t - 6.00 \times 10^{-5} \cos 2t$

C and $I \approx -e^{-25t} (0.0120 \cos 96.82t + 0.0031 \sin 96.82t) + 0.0120 \cos 2t + 0.00012 \sin 2t$ A.

Question 4: Numerical Differentiation

(a) Rate of population growth

Question 4(a). The table below gives the census population (in millions) of a state for the years 1961 to 2001. Find the rate of growth of the population in the year 2001. (6 marks)

Year (x)	1961	1971	1981	1991	2001
Population (y)	19.96	36.65	58.81	77.21	94.61

The year 2001 is at the **end** of the table, so we use Newton's backward difference formula with $h=10$ years.

Backward difference table.

Year x	y	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$
1961	19.96				
1971	36.65	16.69			
1981	58.81	22.16	5.47		
1991	77.21	18.40	-3.76	-9.23	
2001	94.61	17.40	-1.00	2.76	11.99

Derivative at the last point $x_n = 2001$ (where $p=0$):

$$\left(\frac{dy}{dx} \right)_{x_n} = \frac{1}{h} \left[\nabla y_n + \frac{1}{2} \nabla^2 y_n + \frac{1}{3} \nabla^3 y_n + \frac{1}{4} \nabla^4 y_n \right]$$

$$\therefore \frac{1}{10} \left[17.40 + \frac{-1.00}{2} + \frac{2.76}{3} + \frac{11.99}{4} \right] = \frac{1}{10} [17.40 - 0.50 + 0.92 + 2.9975] = \frac{20.8175}{10} = 2.0818$$

Answer. In 2001 the population was growing at about **2.08 million per year**.

(b) Derivatives from tabulated data

Question 4(b). Given the following data points, determine $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ at $x=1.1$ and $x=1.5$. (6 marks)

x	1.0	1.1	1.2	1.3	1.4	1.5	1.6
$f(x)$	7.989	8.403	8.781	9.129	9.451	9.750	10.031

Here $h=0.1$. The point $x=1.1$ is near the start of the table, so we use Newton's **forward** formula there; $x=1.5$ is near the end, so we use the **backward** formula.

Difference table (row x_i lists $\Delta^k f_i$, the forward differences starting at that row).

x	$f(x)$	Δ	Δ^2	Δ^3	Δ^4	Δ^5	Δ^6
1.0	7.989	0.414	-0.036	0.006	-0.002	0.001	0.002
1.1	8.403	0.378	-0.030	0.004	-0.001	0.003	
1.2	8.781	0.348	-0.026	0.003	0.002		
1.3	9.129	0.322	-0.023	0.005			
1.4	9.451	0.299	-0.018				
1.5	9.750	0.281					
1.6	10.031						

At $x=1.1$ (forward differences, $x_0=1.1$)

From the table, $\Delta y_0=0.378$, $\Delta^2 y_0=-0.030$, $\Delta^3 y_0=0.004$, $\Delta^4 y_0=-0.001$ and $\Delta^5 y_0=0.003$

$$\left(\frac{dy}{dx}\right)_{x_0} = \frac{1}{h} \left[\Delta y_0 - \frac{1}{2} \Delta^2 y_0 + \frac{1}{3} \Delta^3 y_0 - \frac{1}{4} \Delta^4 y_0 + \frac{1}{5} \Delta^5 y_0 \right]$$

$$\therefore \frac{1}{0.1} \left[0.378 + \frac{0.030}{2} + \frac{0.004}{3} + \frac{0.001}{4} + \frac{0.003}{5} \right] = 10 [0.378 + 0.015 + 0.00133 + 0.00025 + 0.0006] = 3.952$$

$$\left(\frac{d^2y}{dx^2}\right)_{x_0} = \frac{1}{h^2} \left[\Delta^2 y_0 - \Delta^3 y_0 + \frac{11}{12} \Delta^4 y_0 - \frac{5}{6} \Delta^5 y_0 \right]$$

$$\therefore 100 \left[-0.030 - 0.004 + \frac{11}{12} (-0.001) - \frac{5}{6} (0.003) \right] = 100 [-0.030 - 0.004 - 0.00092 - 0.0025] = -3.742$$

At $x=1.5$ (backward differences, $x_n=1.5$)

From the table, $\nabla y_n=0.299$, $\nabla^2 y_n=-0.023$, $\nabla^3 y_n=0.003$, $\nabla^4 y_n=-0.001$ and $\nabla^5 y_n=0.001$.

$$\left(\frac{dy}{dx}\right)_{x_n} = \frac{1}{h} \left[\nabla y_n + \frac{1}{2} \nabla^2 y_n + \frac{1}{3} \nabla^3 y_n + \frac{1}{4} \nabla^4 y_n + \frac{1}{5} \nabla^5 y_n \right]$$

$$\therefore 10 [0.299 - 0.0115 + 0.001 - 0.00025 + 0.0002] = 2.885$$

$$\left(\frac{d^2y}{dx^2}\right)_{x_n} = \frac{1}{h^2} \left[\nabla^2 y_n + \nabla^3 y_n + \frac{11}{12} \nabla^4 y_n + \frac{5}{6} \nabla^5 y_n \right]$$

$$\hookrightarrow 100[-0.023+0.003-0.00092+0.00083]=-2.008$$

Answer. At $x=1.1$: $\frac{dy}{dx} \approx 3.952$ and $\frac{d^2y}{dx^2} \approx -3.742$. At $x=1.5$: $\frac{dy}{dx} \approx 2.885$ and $\frac{d^2y}{dx^2} \approx -2.008$.

Question 5: Eigenvalues and Eigenvectors

(a) Eigenvalues and eigenvectors of a 3×3 matrix

Question 5(a). Find the eigenvalues and eigenvectors of the matrix (6 marks)

$$A = \begin{bmatrix} 4 & -4 & 3 \\ 0 & -1 & 9 \\ 0 & 0 & 1 \end{bmatrix}$$

Eigenvalues. A is upper triangular, so

$$\det(A - \lambda I) = (4 - \lambda)(-1 - \lambda)(1 - \lambda) = 0 \Rightarrow \lambda = 4, -1, 1$$

(Expanded: $\lambda^3 - 4\lambda^2 - \lambda + 4 = 0$.)

Eigenvectors.

$$\lambda = 4: \begin{bmatrix} 0 & -4 & 3 \\ 0 & -5 & 9 \\ 0 & 0 & -3 \end{bmatrix} x = 0 \Rightarrow x_3 = 0, x_2 = 0, x_1 \text{ free} \Rightarrow x = (1, 0, 0)^T$$

$$\lambda = -1: \begin{bmatrix} 5 & -4 & 3 \\ 0 & 0 & 9 \\ 0 & 0 & 2 \end{bmatrix} x = 0 \Rightarrow x_3 = 0, 5x_1 = 4x_2 \Rightarrow x = (4, 5, 0)^T$$

$$\lambda = 1: \begin{bmatrix} 3 & -4 & 3 \\ 0 & -2 & 9 \\ 0 & 0 & 0 \end{bmatrix} x = 0 \Rightarrow x_2 = 9/2 x_3, 3x_1 = 4x_2 - 3x_3 = 15x_3 \Rightarrow x_1 = 5x_3. \text{ Taking } x_3 = 2:$$

$$x = (10, 9, 2)^T$$

Check for $\lambda = 1$: $A(10, 9, 2)^T = (40 - 36 + 6, -9 + 18, 2)^T = (10, 9, 2)^T \checkmark$

Answer. $\lambda_1 = 4$ with $x_1 = k(1, 0, 0)^T$; $\lambda_2 = -1$ with $x_2 = k(4, 5, 0)^T$; $\lambda_3 = 1$ with $x_3 = k(10, 9, 2)^T$ ($k \neq 0$).

(b) Principal directions of a stretched membrane

Question 5(b). An elastic membrane in the x_1x_2 -plane with boundary circle $x_1^2 + x_2^2 = 1$ is stretched so that a point $P:(x_1, x_2)$ goes over into the point $Q:(y_1, y_2)$ given by

$$y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = A x = \begin{bmatrix} 5 & 3 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Find the principal directions, that is, the directions of the position vector x of P for which the direction of the position vector y of Q is the same or exactly opposite. (6 marks)

We need the vectors x for which $y = Ax$ is parallel to x , that is $Ax = \lambda x$. These are the eigenvectors of

$$A = \begin{bmatrix} 5 & 3 \\ 3 & 5 \end{bmatrix}$$

Eigenvalues.

$$\det(A - \lambda I) = (5 - \lambda)^2 - 9 = \lambda^2 - 10\lambda + 16 = (\lambda - 8)(\lambda - 2) = 0 \Rightarrow \lambda_1 = 8, \lambda_2 = 2$$

Eigenvectors.

For $\lambda_1 = 8$: $-3x_1 + 3x_2 = 0 \Rightarrow x_2 = x_1$, so $x^{(1)} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, at 45° to the x_1 -axis.

For $\lambda_2 = 2$: $3x_1 + 3x_2 = 0 \Rightarrow x_2 = -x_1$, so $x^{(2)} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, at 135° (or -45°) to the x_1 -axis.

Both eigenvalues are positive, so in each principal direction y points the **same** way as x . The two directions are perpendicular because A is symmetric.

Interpretation. Along the 45° direction the membrane is stretched by a factor of 8, and along the 135° direction by a factor of 2. With coordinates u_1, u_2 along these principal directions, the boundary circle $x_1^2 + x_2^2 = 1$ becomes the ellipse

$$\frac{u_1^2}{8^2} + \frac{u_2^2}{2^2} = 1$$

Answer. The principal directions are $x_2 = x_1$ (at 45° , eigenvalue 8, stretch factor 8) and $x_2 = -x_1$ (at 135° , eigenvalue 2, stretch factor 2). The circle is deformed into an ellipse with semi-axes 8 and 2 along these directions.

Question 6: Series Solutions and Bessel's Equation

(a) Power series solution

Question 6(a). Find the power series solution of $5(y'' + y) = 0$. (6 marks)

Dividing by the non-zero constant 5 leaves $y'' + y = 0$.

Assume a power series solution about $x=0$:

$$y = \sum_{n=0}^{\infty} a_n x^n, y'' = \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} = \sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2} x^n$$

Substituting into $y'' + y = 0$:

$$\sum_{n=0}^{\infty} [(n+2)(n+1)a_{n+2} + a_n] x^n = 0$$

Every coefficient must vanish, which gives the **recurrence relation**

$$a_{n+2} = -\frac{a_n}{(n+2)(n+1)}, n=0, 1, 2, \dots$$

Even coefficients (in terms of a_0):

$$a_2 = -\frac{a_0}{2!}, a_4 = -\frac{a_2}{4 \cdot 3} = \frac{a_0}{4!}, a_6 = -\frac{a_4}{6 \cdot 5} = -\frac{a_0}{6!}, \dots$$

Odd coefficients (in terms of a_1):

$$a_3 = -\frac{a_1}{3!}, a_5 = -\frac{a_3}{5 \cdot 4} = \frac{a_1}{5!}, a_7 = -\frac{a_1}{7!}, \dots$$

Hence

$$y = a_0 \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots \right) + a_1 \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots \right)$$

The two series are the Maclaurin series of $\cos x$ and $\sin x$, and both converge for all x .

Answer. $y = a_0 \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots \right) + a_1 \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots \right) = a_0 \cos x + a_1 \sin x$, where a_0 and a_1 are arbitrary constants.

(b) Bessel's equation

Question 6(b). Find the general solution, in terms of Bessel functions, of the equation $x y'' + y' + \left(\frac{5}{10}\right) y = 0$. (6 marks)

The fraction in the equation reduces to $1/2$, so the equation is

$$x y'' + y' + 1/2 y = 0$$

Change of variable. Equations of the form $x y'' + y' + \lambda y = 0$ become Bessel's equation under $z = 2\sqrt{\lambda x}$. Here $\lambda = 1/2$, so put

$$z = 2\sqrt{x/2} = \sqrt{2x}, z^2 = 2x, \frac{dz}{dx} = \frac{1}{z}$$

Then

$$\frac{dy}{dx} = \frac{1}{z} \frac{dy}{dz}, \frac{d^2y}{dx^2} = \frac{1}{z} \frac{d}{dz} \left(\frac{1}{z} \frac{dy}{dz} \right) = \frac{1}{z^2} \frac{d^2y}{dz^2} - \frac{1}{z^3} \frac{dy}{dz}$$

With $x = z^2/2$:

$$x \frac{d^2 y}{dx^2} + \frac{dy}{dx} + \frac{1}{2}y = \frac{1}{2} \frac{d^2 y}{dz^2} - \frac{1}{2z} \frac{dy}{dz} + \frac{1}{z} \frac{dy}{dz} + \frac{1}{2}y = \frac{1}{2} \left(\frac{d^2 y}{dz^2} + \frac{1}{z} \frac{dy}{dz} + y \right) = 0$$

Multiplying by $2z^2$:

$$z^2 \frac{d^2 y}{dz^2} + z \frac{dy}{dz} + (z^2 - \nu^2)y = 0 \text{ with } \nu = 0$$

This is **Bessel's equation of order** $\nu = 0$.

General solution. For order $\nu = 0$ the two independent solutions are $J_0(z)$ and the Bessel function of the second kind, $Y_0(z)$:

$$y = c_1 J_0(z) + c_2 Y_0(z) = c_1 J_0(\sqrt{2x}) + c_2 Y_0(\sqrt{2x})$$

In series form, the first solution is

$$J_0(\sqrt{2x}) = \sum_{m=0}^{\infty} \frac{(-1)^m}{(m!)^2} \left(\frac{x}{2} \right)^m = 1 - \frac{x}{2} + \frac{x^2}{16} - \frac{x^3}{288} + \dots$$

Answer. $y = c_1 J_0(\sqrt{2x}) + c_2 Y_0(\sqrt{2x})$.

Question 7: Legendre's Equation and the Wave Equation

(a) Legendre's equation

Question 7(a). Solve this special Legendre equation, $(1-x^2)y'' - 2xy' + 2y = 0$, which occurs in models exhibiting spherical symmetry. (6 marks)

This is Legendre's equation $(1-x^2)y'' - 2xy' + n(n+1)y = 0$ with $n(n+1) = 2$, that is $n = 1$.

Power series. Put $y = \sum_{m=0}^{\infty} a_m x^m$. Then

$$(1-x^2) \sum_m m(m-1)a_m x^{m-2} - 2x \sum_m m a_m x^{m-1} + 2 \sum_m a_m x^m = 0$$

Collecting the coefficient of x^m :

$$(m+2)(m+1)a_{m+2} - [m(m-1) + 2m - 2]a_m = 0$$

Since $m(m-1) + 2m - 2 = m^2 + m - 2 = (m+2)(m-1)$, the recurrence relation is

$$a_{m+2} = \frac{(m+2)(m-1)}{(m+2)(m+1)} a_m = \frac{m-1}{m+1} a_m$$

Odd series ($m=1$): $a_3 = 0 \cdot a_1 = 0$, so $a_5 = a_7 = \dots = 0$ and the series stops:

$$y_1 = a_1 x = a_1 P_1(x)$$

Even series (from a_0):

$$a_2 = -a_0, a_4 = \frac{1}{3}a_2 = -\frac{a_0}{3}, a_6 = \frac{3}{5}a_4 = -\frac{a_0}{5}, a_8 = -\frac{a_0}{7}, \dots$$

$$y_2 = a_0 \left(1 - x^2 - \frac{x^4}{3} - \frac{x^6}{5} - \frac{x^8}{7} - \dots \right) = a_0 \left(1 - \frac{x}{2} \ln \frac{1+x}{1-x} \right), \forall x \in (-1, 1)$$

because $\frac{1}{2} \ln \frac{1+x}{1-x} = x + \frac{x^3}{3} + \frac{x^5}{5} + \dots$.

Check of $y_1 = x$: $(1-x^2)(0) - 2x(1) + 2x = 0$. ✓

Answer. $y = c_1 x + c_2 \left(1 - x^2 - \frac{x^4}{3} - \frac{x^6}{5} - \dots \right) = c_1 P_1(x) + c_2 \left(1 - \frac{x}{2} \ln \frac{1+x}{1-x} \right)$ for $|x| < 1$. The polynomial solution is the Legendre polynomial $P_1(x) = x$.

(b) One-dimensional wave equation

Question 7(b). Find the solution to this one-dimensional wave equation:

$\frac{\partial^2 u}{\partial x^2} = \frac{1}{16} \frac{\partial^2 u}{\partial t^2}$ for $0 < x < 2, t > 0$. The boundary conditions are $u(0, t) = u(2, t) = 0$. The initial conditions are (i) $u(x, 0) = 6 \sin \pi x - 3 \sin 4 \pi x$ and (ii) $\frac{\partial u}{\partial t}(x, 0) = 0$. (6 marks)

The equation is

$$\frac{\partial^2 u}{\partial x^2} = \frac{1}{16} \frac{\partial^2 u}{\partial t^2} \Leftrightarrow \frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}, c^2 = 16, c = 4, L = 2$$

Separation of variables. Put $u = F(x)G(t)$:

$$\frac{F''}{F} = \frac{G''}{16G} = -p^2$$

so $F'' + p^2 F = 0$ and $G'' + 16p^2 G = 0$.

Boundary conditions. $u(0, t) = u(2, t) = 0$ need $F(0) = F(2) = 0$. With $F = A \cos p x + B \sin p x$, $F(0) = 0$ gives $A = 0$, and $\sin 2p = 0$ gives

$$p = \frac{n\pi}{2}, F_n(x) = \sin \frac{n\pi x}{2}, n = 1, 2, 3, \dots$$

Time part. $G_n'' + \lambda_n^2 G_n = 0$ with $\lambda_n = \frac{cn\pi}{L} = \frac{4n\pi}{2} = 2n\pi$:

$$G_n(t) = B_n \cos 2n\pi t + B_n^i \sin 2n\pi t$$

so

$$u(x, t) = \sum_{n=1}^{\infty} \left(B_n \cos 2n\pi t + B_n^i \sin 2n\pi t \right) \sin \frac{n\pi x}{2}$$

Initial velocity. $\frac{\partial u}{\partial t}(x, 0) = \sum 2n\pi B_n^i \sin \frac{n\pi x}{2} = 0$, so $B_n^i = 0$ for every n .

Initial displacement.

$$u(x, 0) = \sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{2} = 6 \sin \pi x - 3 \sin 4\pi x$$

$\sin \pi x = \sin \frac{2\pi x}{2}$ is the $n=2$ term and $\sin 4\pi x = \sin \frac{8\pi x}{2}$ is the $n=8$ term. Comparing coefficients (no Fourier integrals are needed):

$$B_2 = 6, B_8 = -3, B_n = 0 \text{ otherwise}$$

With $\lambda_2 = 4\pi$ and $\lambda_8 = 16\pi$:

| **Answer.** $u(x, t) = 6 \sin \pi x \cos 4\pi t - 3 \sin 4\pi x \cos 16\pi t$.

Check. $u(0, t) = u(2, t) = 0$ since $\sin 2\pi = \sin 8\pi = 0$; $u(x, 0) = 6 \sin \pi x - 3 \sin 4\pi x$; $u_t(x, 0) = 0$; and each term satisfies $u_{tt} = 16u_{xx}$, for example $(4\pi)^2 = 16\pi^2$. ✓